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Malware Analysis Report: Agent Tesla

For this report, I chose to go with a popular strain of malware that seems to appear nearly everywhere on the MalwareBazaar Database. I suspect that this might be due to its source code either being released or leaked to the public. Either way, Agent Tesla is categorized as a RAT and since new variations of it are popping up quite fast, I decided that it would be an interesting piece of malware to learn about (abuse.ch).

The specific strain of Agent Tesla we are going to be analyzing is identified by its SHA-256 hash: 2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2

Links to references will be provided at the end of the report.

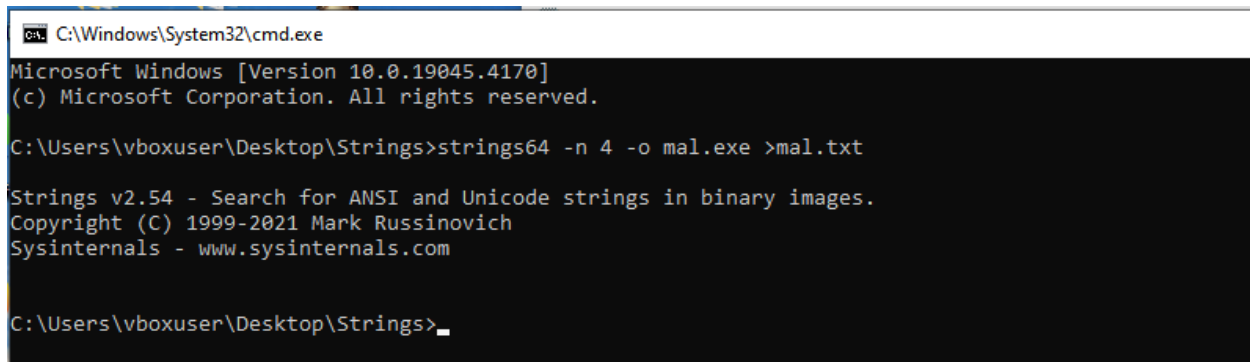
This report will include the usage of a Windows 10 Virtual Machine and the following tools: Strings64, UPX (did not work), PEview, VirusTotal, Regshot, Process Monitor, ApatеDNS, Wireshark, x64 Debugger, and IDA pro.

Tools not used in this report: Resource Hacker (was not needed), and Dependency walker (had too many errors).

AgentTesla: Strings

First, we will utilize Strings to get a sense of what functions and literal strings are present in the executable.

Figure 1



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vboxuser\Desktop\Strings>strings64 -n 4 -o mal.exe >mal.txt

Strings v2.54 - Search for ANSI and Unicode strings in binary images.
Copyright (C) 1999-2021 Mark Russinovich
Sysinternals - www.sysinternals.com

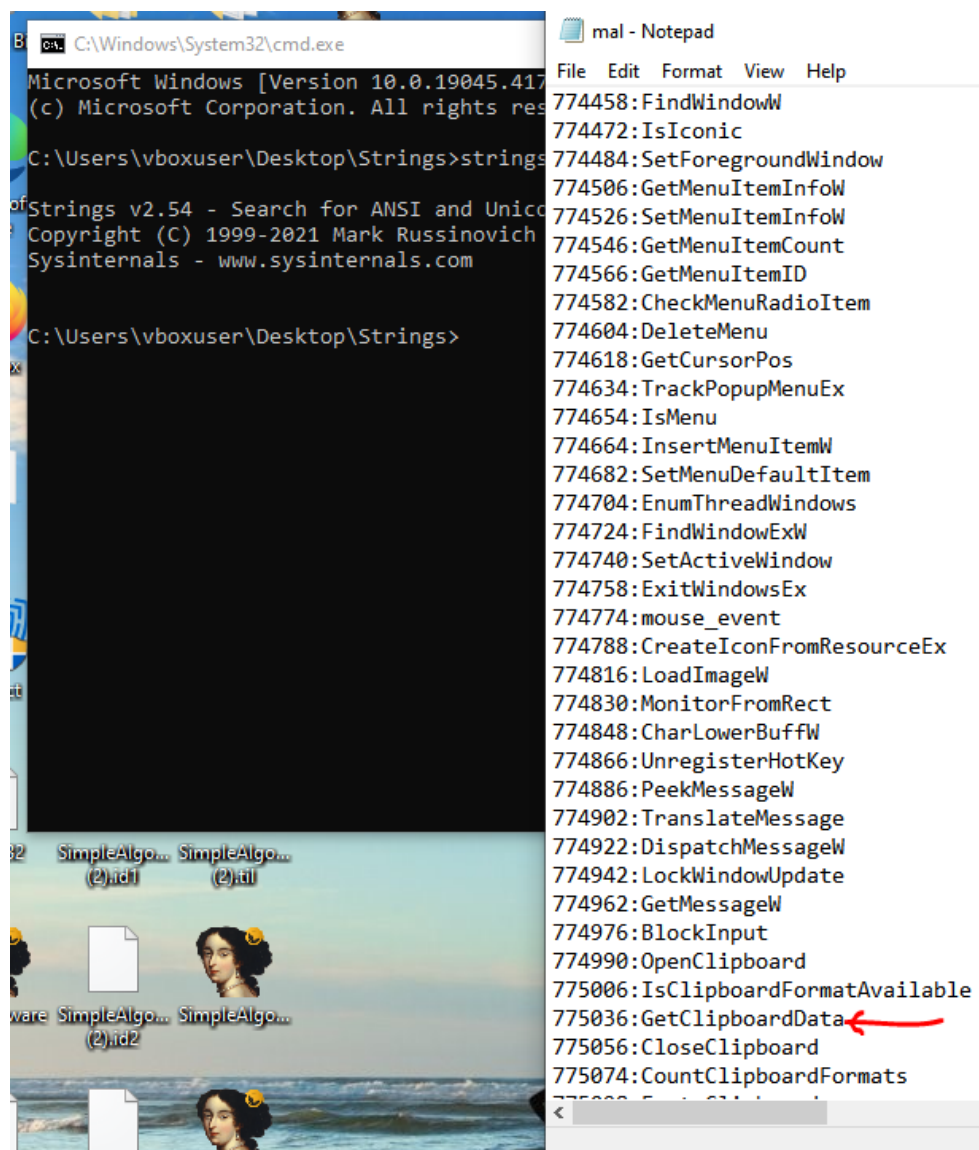
C:\Users\vboxuser\Desktop\Strings>_
```

Upon successfully running Strings, we are presented with a massive text file containing around 1,000,000 lines of text. Most of which I ended up erasing because it was undecipherable. After doing so, I ended up with around a few thousand lines of text. This executable has a whole myriad of functions that expand its capabilities. While going through some of the functions I found a few to be noteworthy:

775036: GetClipboardData

This function is fairly self-explanatory, it allows the executable to read whatever is stored in the user's clipboard. Seeing how we are dealing with a RAT; this could be used maliciously to extract some sensitive information a user does not wish to type out. This is especially important since RATs usually come with keyloggers, making this RAT very versatile.

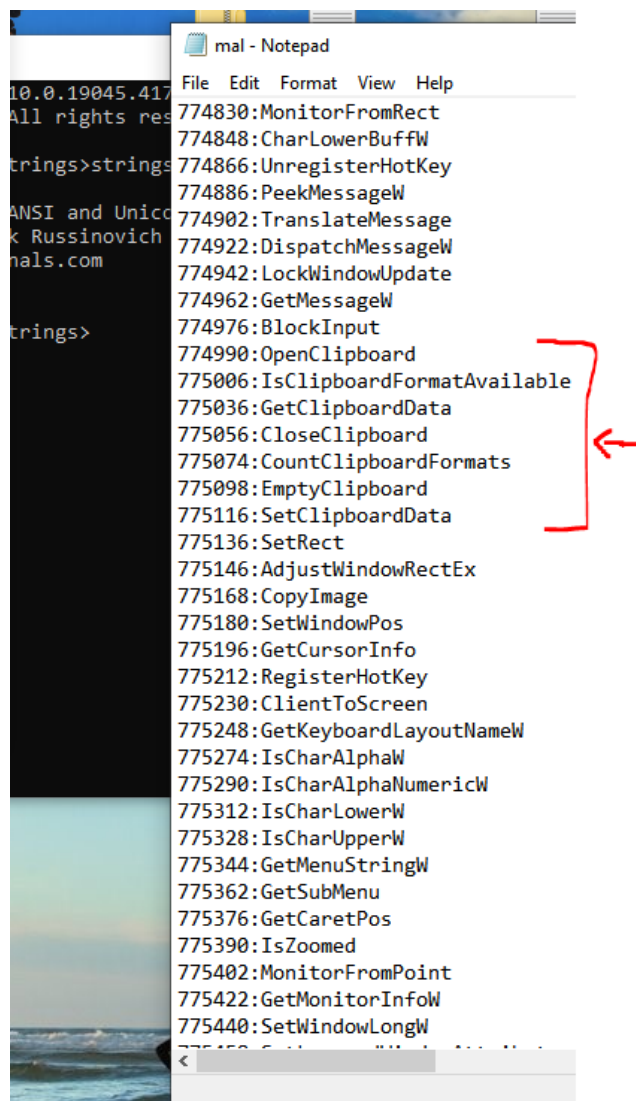
Figure 2.



774990: OpenClipboard
775006: IsClipboardFormatAvailable
775036: GetClipboardData
775056: CloseClipboard
775074: CountClipboardFormats
775098: EmptyClipboard
775116: SetClipboardData

As we can see, similar to the previous figure we have a cluster of functions that all directly interact with our clipboard, even as far as emptying and setting our clipboard data, being highly invasive.

Figure 3.



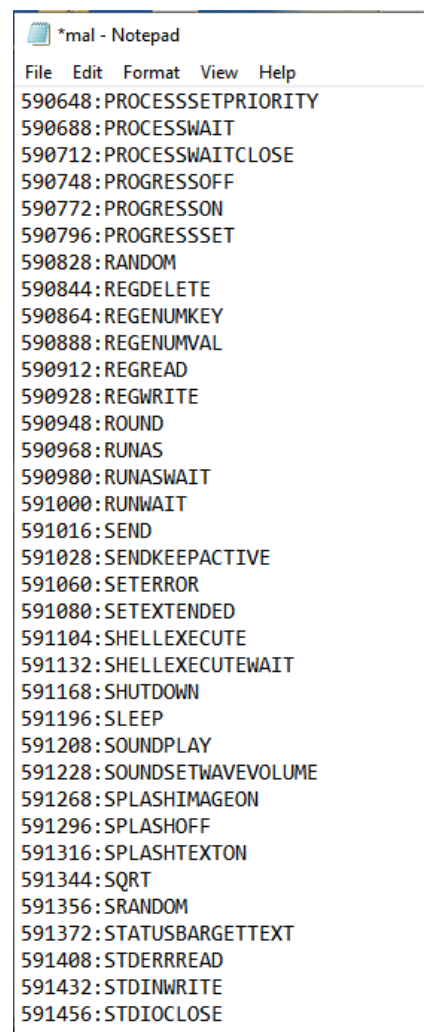
More Highly Invasive Functions:

While I can't provide the whole list in an image, I will provide an extra two photos. Keep in mind that there are thousands of functions similar to the ones in Figures 4 and 5. Indicating that this RAT is not meant to be compact and precise. If infected, it should grant the attacker full unfettered access to a victim's PC. Some dangerous functions include: FILECOPY, FILEDELETE, FILEMOVE, SHELLEXECUTE, SOUNDPLAY, SHUTDOWN, etc...



```
*mal - Notepad
File Edit Format View Help
585648:[>:]
585656:[<:]
585666:FILECLOSE
585688:FILECOPY
585708:FILECREATENTFSLINK
585748:FILECREATESHORTCUT
585788:FILEDELETE
585812:FILEEXISTS
585836:FILEFINDFIRSTFILE
585872:FILEFINDNEXTFILE
585908:FILEFLUSH
585928:FILEGETATTRIB
585956:FILEGETENCODING
585988:FILEGETLONGNAME
586020:FILEGETPOS
586044:FILEGETSHORTCUT
586076:FILEGETSHORTNAME
586112:FILEGETSIZE
586136:FILEGETTIME
586160:FILEGETVERSION
586192:FILEINSTALL
586216:FILEMOVE
586236:FILEOPEN
586256:FILEOPENDIALOG
586288:FILEREAD
586308:FILEREADLINE
586336:FILEREADTOARRAY
586368:FILERECYCLE
586392:FILERECYCLEEMPTY
586428:FILESAVEDIALOG
586460:FILESELECTFOLDER
586496:FILESETATTRIB
586524:FILESETEND
586548:FILESETPOS
586572:FILESETTIME
```

Figure 4.



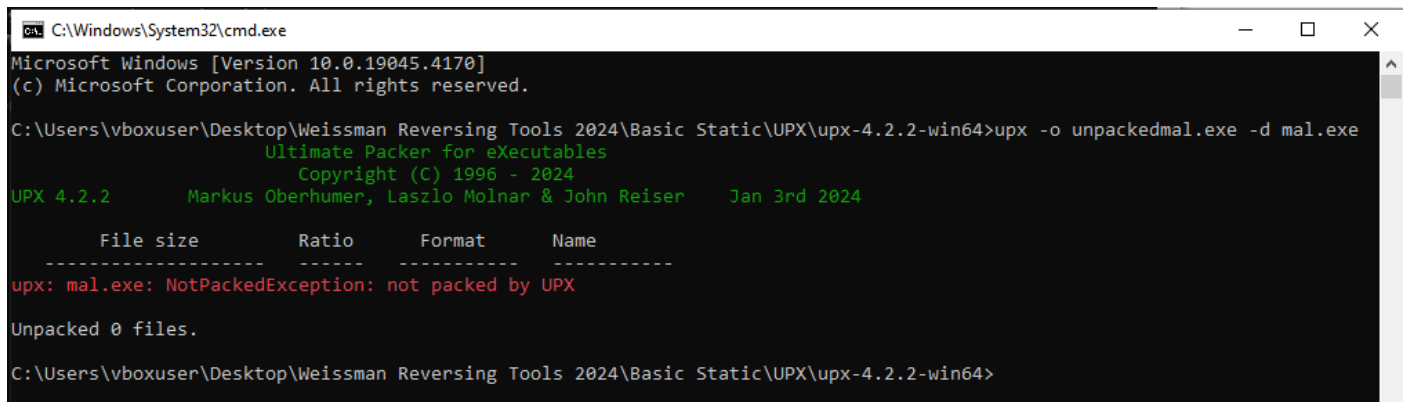
```
*mal - Notepad
File Edit Format View Help
590648:PROCESSSETPRIORITY
590688:PROCESSWAIT
590712:PROCESSWAITCLOSE
590748:PROGRESSOFF
590772:PROGRESSION
590796:PROGRESSSET
590828:RANDOM
590844:REGDELETE
590864:REGENUMKEY
590888:REGENUMVAL
590912:REGREAD
590928:REGWRITE
590948:ROUND
590968:RUNAS
590980:RUNASWAIT
591000:RUNWAIT
591016:SEND
591028:SENDKEEPACTIVE
591060:SETERROR
591080:SETEXTENDED
591104:SHELLEXECUTE
591132:SHELLEXECUTEWAIT
591168:SHUTDOWN
591196:SLEEP
591208:SOUNDPLAY
591228:SOUNDSETWAVEVOLUME
591268:SPLASHIMAGEON
591296:SPLASHOFF
591316:SPLASHTEXTON
591344:SQRT
591356:SRANDOM
591372:STATUSBARGETTEXT
591408:STDERRREAD
591432:STDINWRITE
591456:STDIOCLOSE
```

Figure 5.

AgentTesla: UPX

As previously stated, there were thousands of lines which were undecipherable. So, I decided to see if it was packed, and if UPX could unpack it. Unfortunately, as we can see in Figure 6, UPX could not unpack it which means it was not packed in that way.

Figure 6.



```
C:\Windows\System32\cmd.exe
Microsoft Windows [Version 10.0.19045.4170]
(c) Microsoft Corporation. All rights reserved.

C:\Users\vboxuser\Desktop\Weissman Reversing Tools 2024\Basic Static\UPX\upx-4.2.2-win64>upx -o unpackedmal.exe -d mal.exe
      Ultimate Packer for eXecutables
      Copyright (C) 1996 - 2024
UPX 4.2.2      Markus Oberhumer, Laszlo Molnar & John Reiser      Jan 3rd 2024

  File size      Ratio      Format      Name
  -----
upx: mal.exe: NotPackedException: not packed by UPX

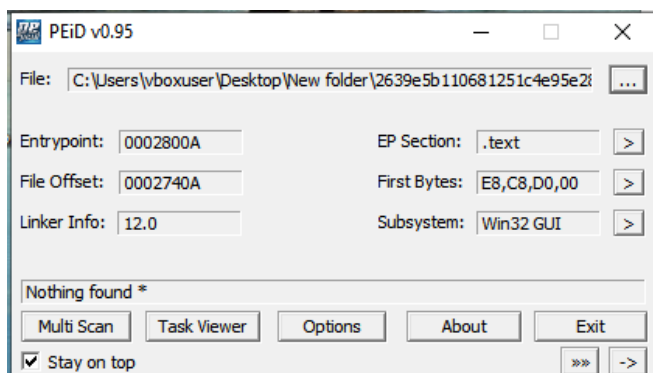
Unpacked 0 files.

C:\Users\vboxuser\Desktop\Weissman Reversing Tools 2024\Basic Static\UPX\upx-4.2.2-win64>
```

AgentTesla: PEiD

Given the results presented by UPX I decided to analyze the file using PEiD. As we can see in Figure 7, this executable does not use any kind of packer (hardcore scanning was enabled here). This means our initial results from Strings are valid, and that the undecipherable text might have been an attempt to obfuscate some data.

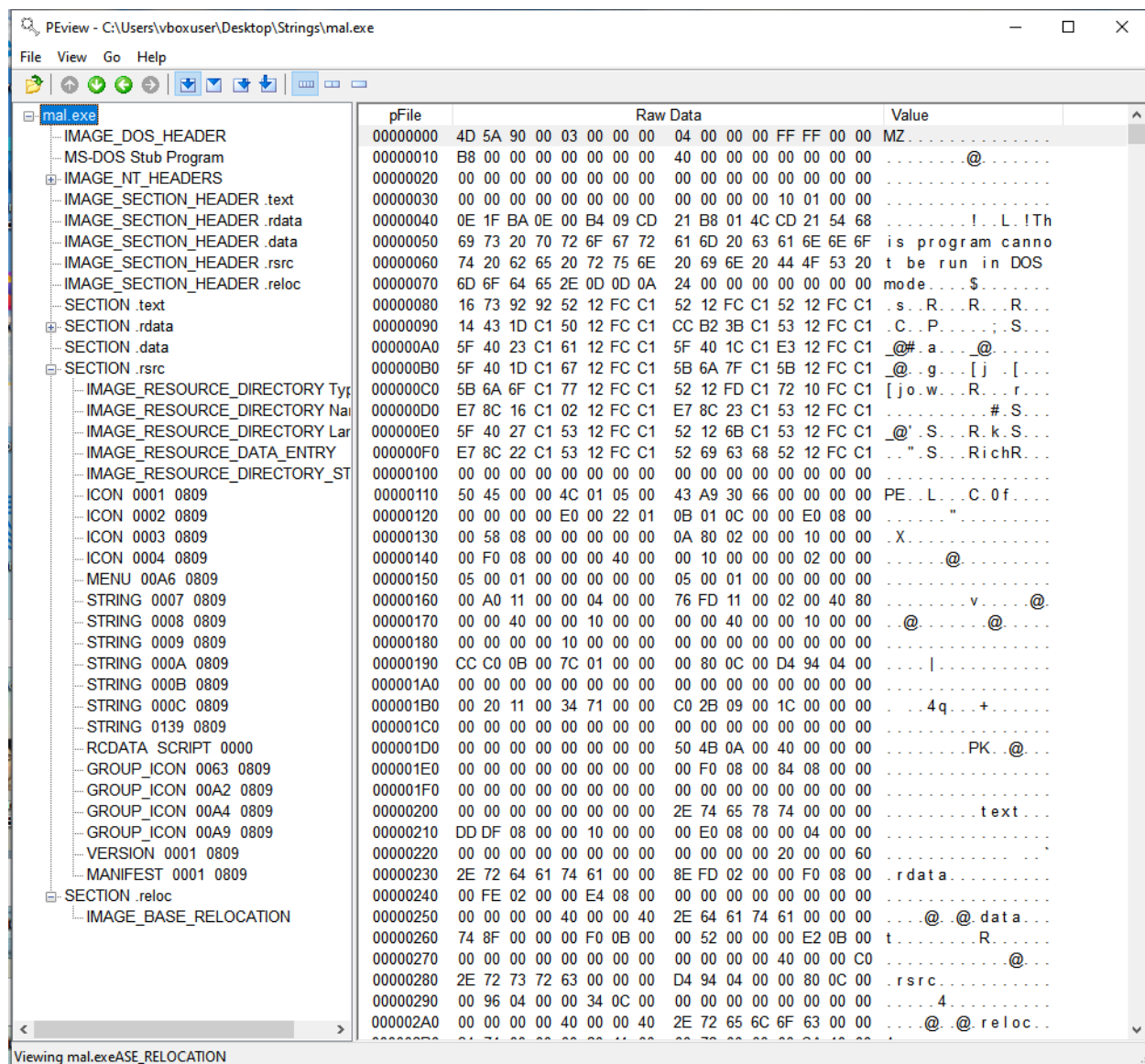
Figure 7.



AgentTesla: PReview

Loading the executable in PReview did not lead to any remarkable intel. However, it did tell us that our executable (a portable executable) is not hiding any other binaries from us. This means that we can somewhat safely assume that most of the functionality of this malware is done directly and solely from this portable executable. We can see this in Figure 8 where no additional binaries are present in the “.rsrc” section. Thus, “resource hacker” was not of use here.

Figure 8.

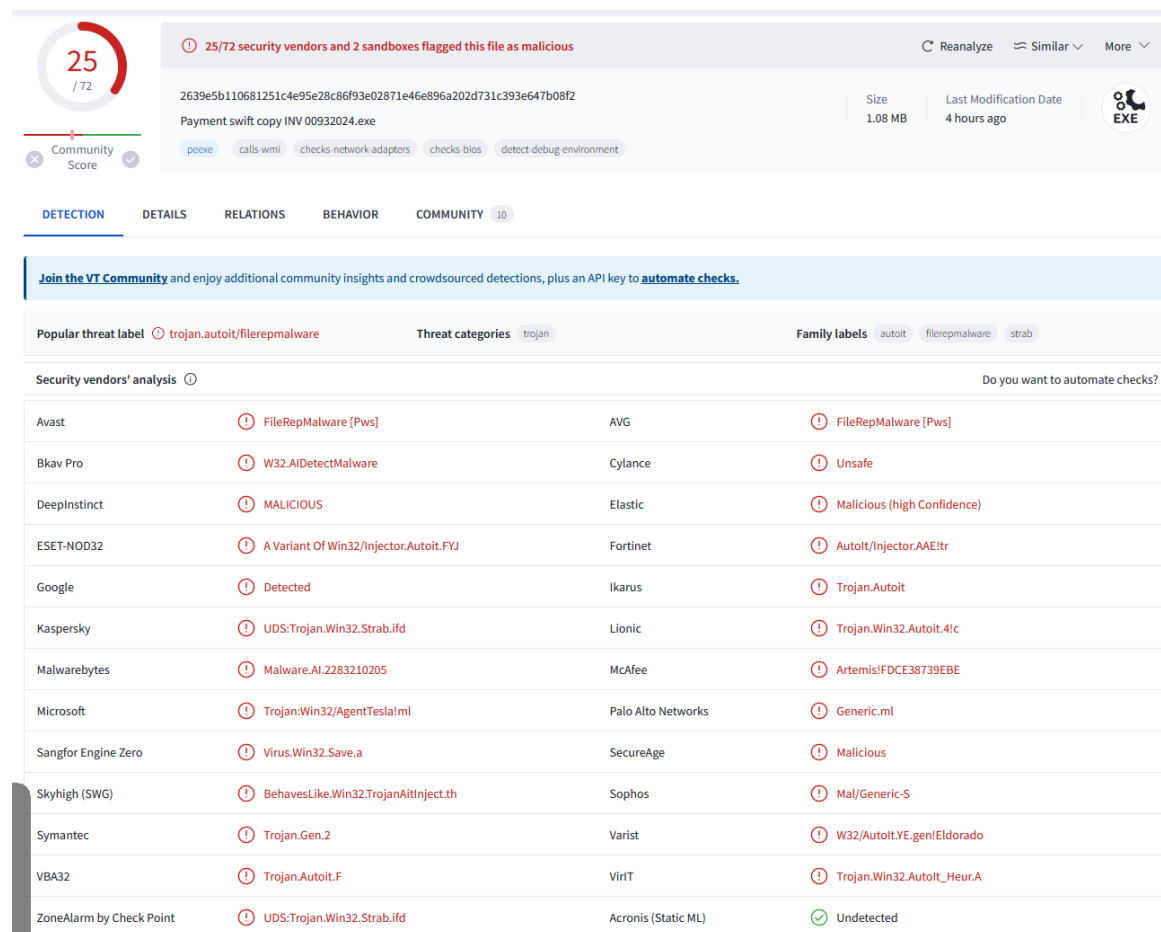


AgentTesla: VirusTotal

VirusTotal was a crucial tool in this analysis. However, later on in the report, we will see that we discover some things that were not yet added to VirusTotal, most likely due to this strain of malware being fairly new. Keeping this in mind, we will do a short return to VirusTotal to go over the unreported findings. Without further ado let's dive into the initial analysis.

According to VirusTotal detection tab in Figure 9, only 25/72 vendors found this executable malicious, this could be partly because this is a newer strain of the Agent Tesla Malware. The good news is that major vendors have caught on to it.

Figure 9.



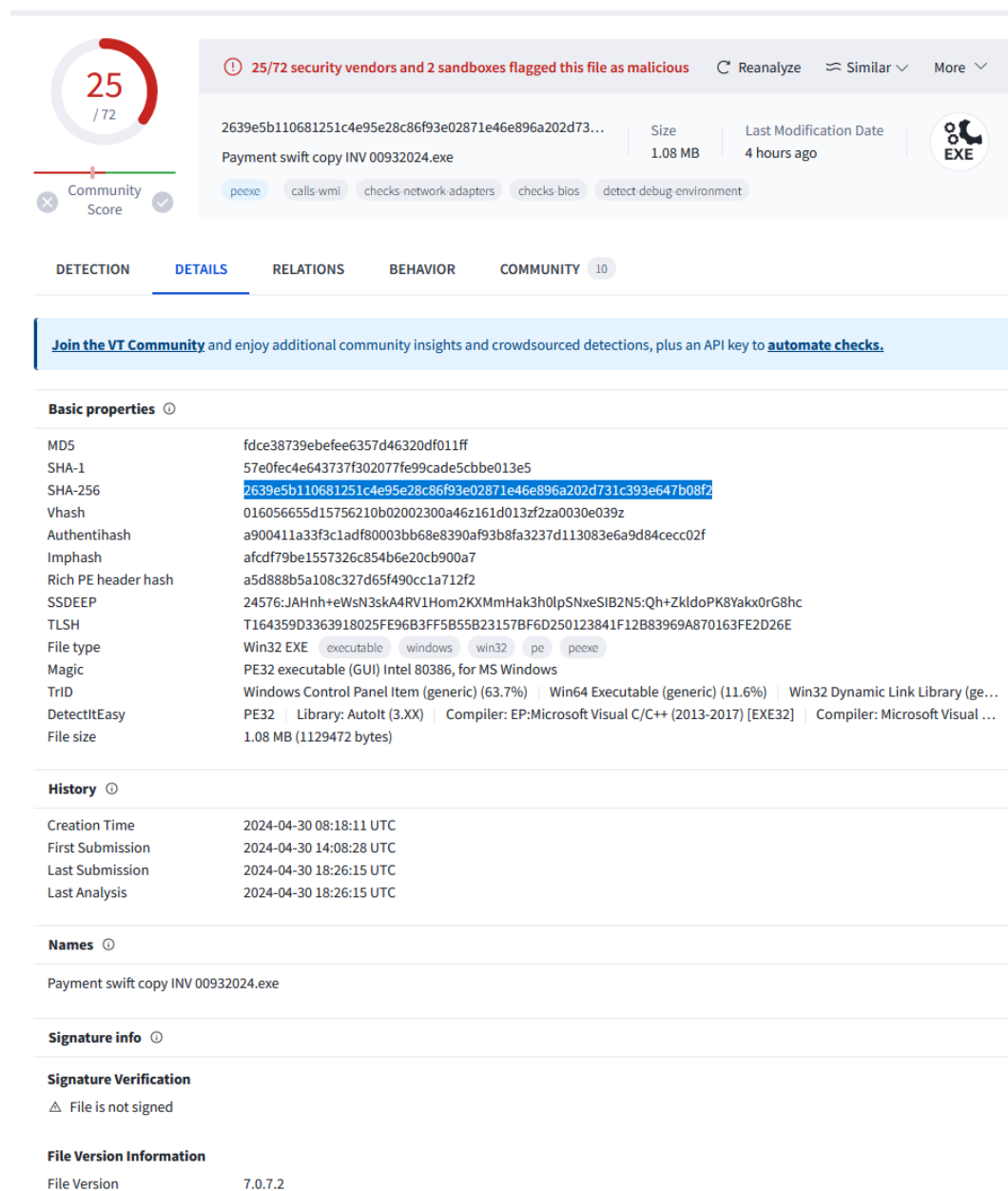
While looking at some of the vendors in Figure 10, I found a few that seem worrying as they did not detect this malware. Bitdefender, Crowdstrike Falcon, VIPRE, and SentinelOne did not detect the malware which is odd since they are major cybersecurity companies. Again, this could be partly because this strain is new.

Figure 10.

ZoneAlarm by Check Point	🚫 UDS:Trojan.Win32.Strab.ifd	Acronis (Static ML)	✅ Undetected
AhnLab-V3	✅ Undetected	Alibaba	✅ Undetected
AliCloud	✅ Undetected	ALYac	✅ Undetected
Antiy-AVL	✅ Undetected	Arcabit	✅ Undetected
Avira (no cloud)	✅ Undetected	Baidu	✅ Undetected
BitDefender	✅ Undetected	BitDefenderTheta	✅ Undetected
ClamAV	✅ Undetected	CMC	✅ Undetected
CrowdStrike Falcon	✅ Undetected	Cynet	✅ Undetected
DrWeb	✅ Undetected	Emsisoft	✅ Undetected
eScan	✅ Undetected	GData	✅ Undetected
Gridinsoft (no cloud)	✅ Undetected	Jiangmin	✅ Undetected
K7AntiVirus	✅ Undetected	K7GW	✅ Undetected
Kingsoft	✅ Undetected	MAX	✅ Undetected
MaxSecure	✅ Undetected	NANO-Antivirus	✅ Undetected
Panda	✅ Undetected	QuickHeal	✅ Undetected
Rising	✅ Undetected	SentinelOne (Static ML)	✅ Undetected
SUPERAntiSpyware	✅ Undetected	TACHYON	✅ Undetected
TEHTRIS	✅ Undetected	Tencent	✅ Undetected
Trapmine	✅ Undetected	Trellix (FireEye)	✅ Undetected
TrendMicro	✅ Undetected	TrendMicro-HouseCall	✅ Undetected
VIPRE	✅ Undetected	ViRobot	✅ Undetected
Webroot	✅ Undetected	WithSecure	✅ Undetected
Xcitium	✅ Undetected	Yandex	✅ Undetected
Zillya	✅ Undetected	Zoner	✅ Undetected

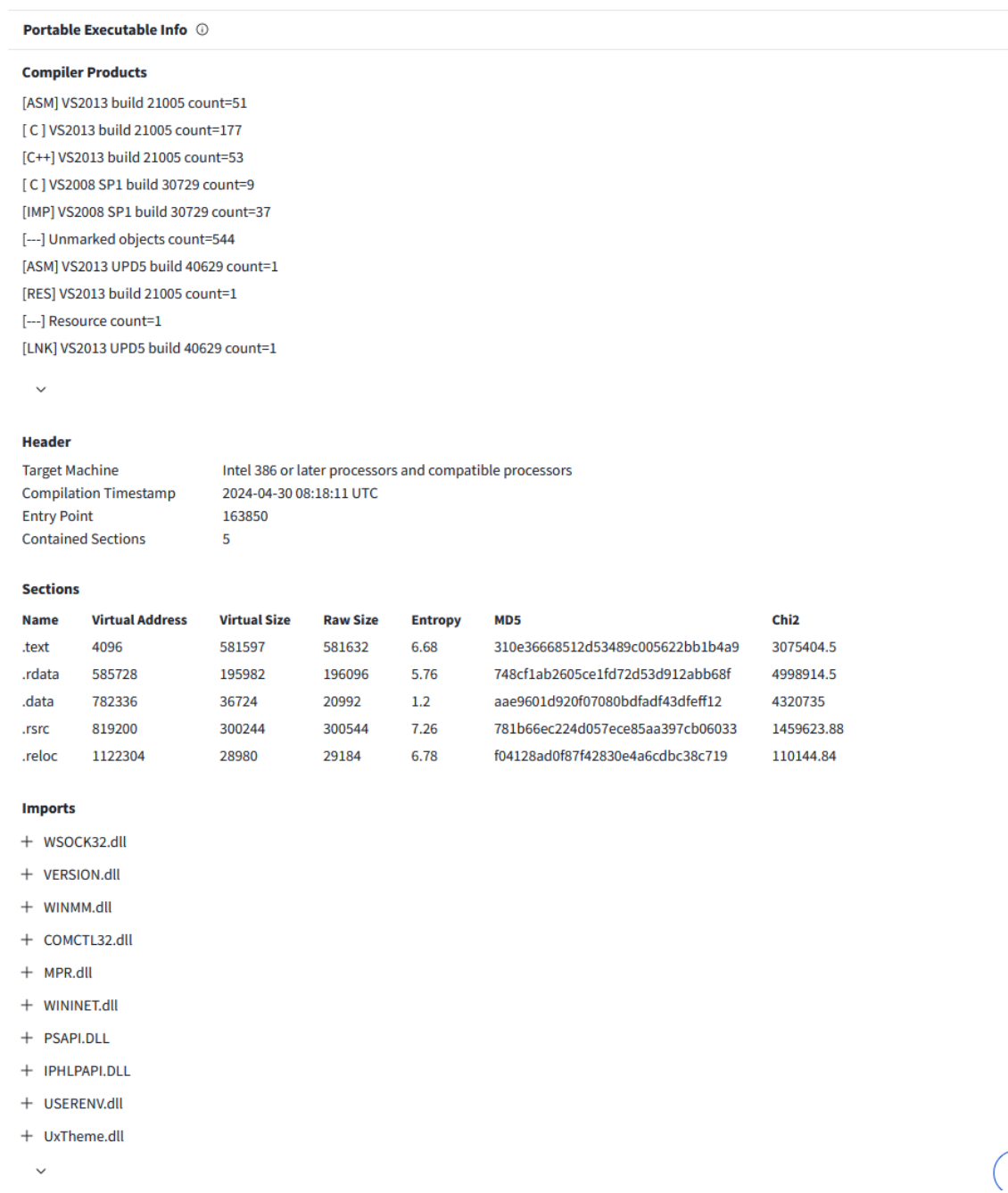
Moving onwards to the details tab in Figure 11, we can see the confirmation that this is indeed the same file we saw on MalwareBazaar as indicated by the matching SHA-256 hash. We also see that this executable was first analyzed by someone using VirusTotal only 4 hours before us! Also, we need to pay attention to the executable name on the top as it might come in handy in the future (INV 00932024.exe).

Figure 11.



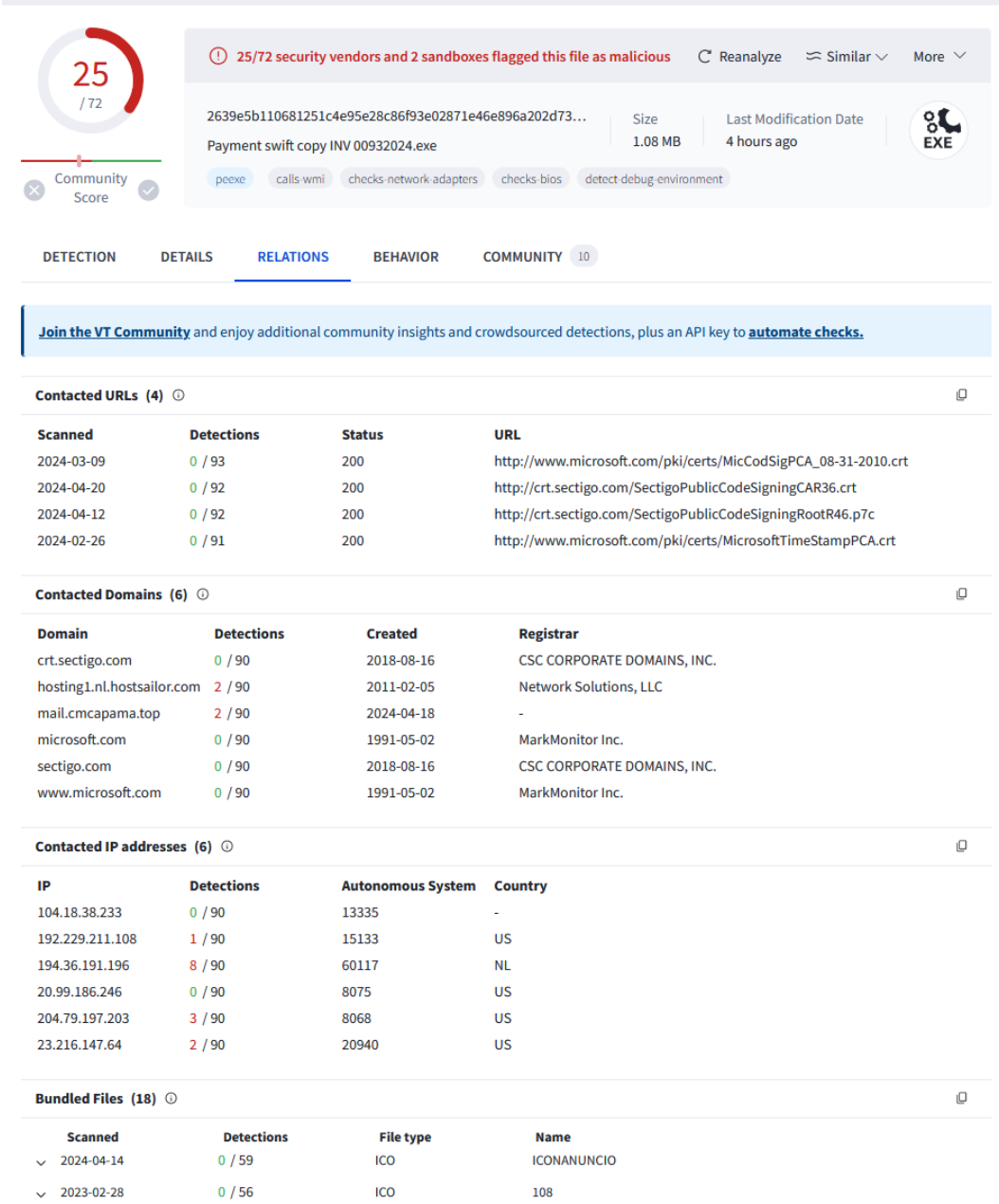
Continuing with the details tab we can see some more relevant information, as mentioned before, some of the information on VirusTotal is incomplete. As such it's worth taking note of the Compiler Products section and its content in Figure 12 as it could have relevance in the secondary analysis.

Figure 12.



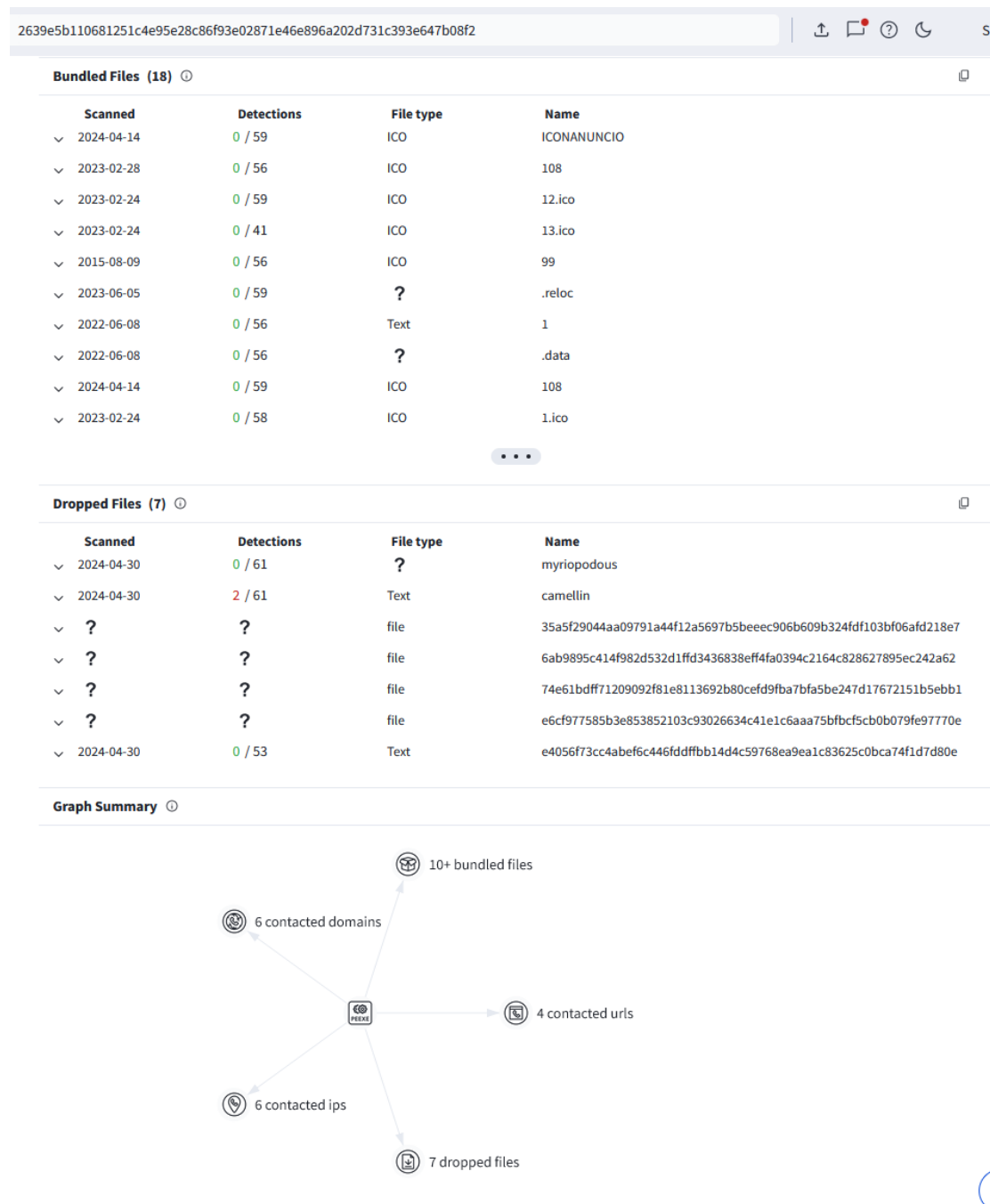
Moving into the relations tab in Figure 13, there is a plethora of information that will be relevant to the secondary analysis. Upon first analysis, we can see that not many of the related IPs or FQDNs are being marked as malicious, again most likely partly due to it being new. It almost looks as if some of these results may be benign. However, note that some of the content presented here (domains) will be referenced in subsequent sections of the report with vindication.

Figure 13.



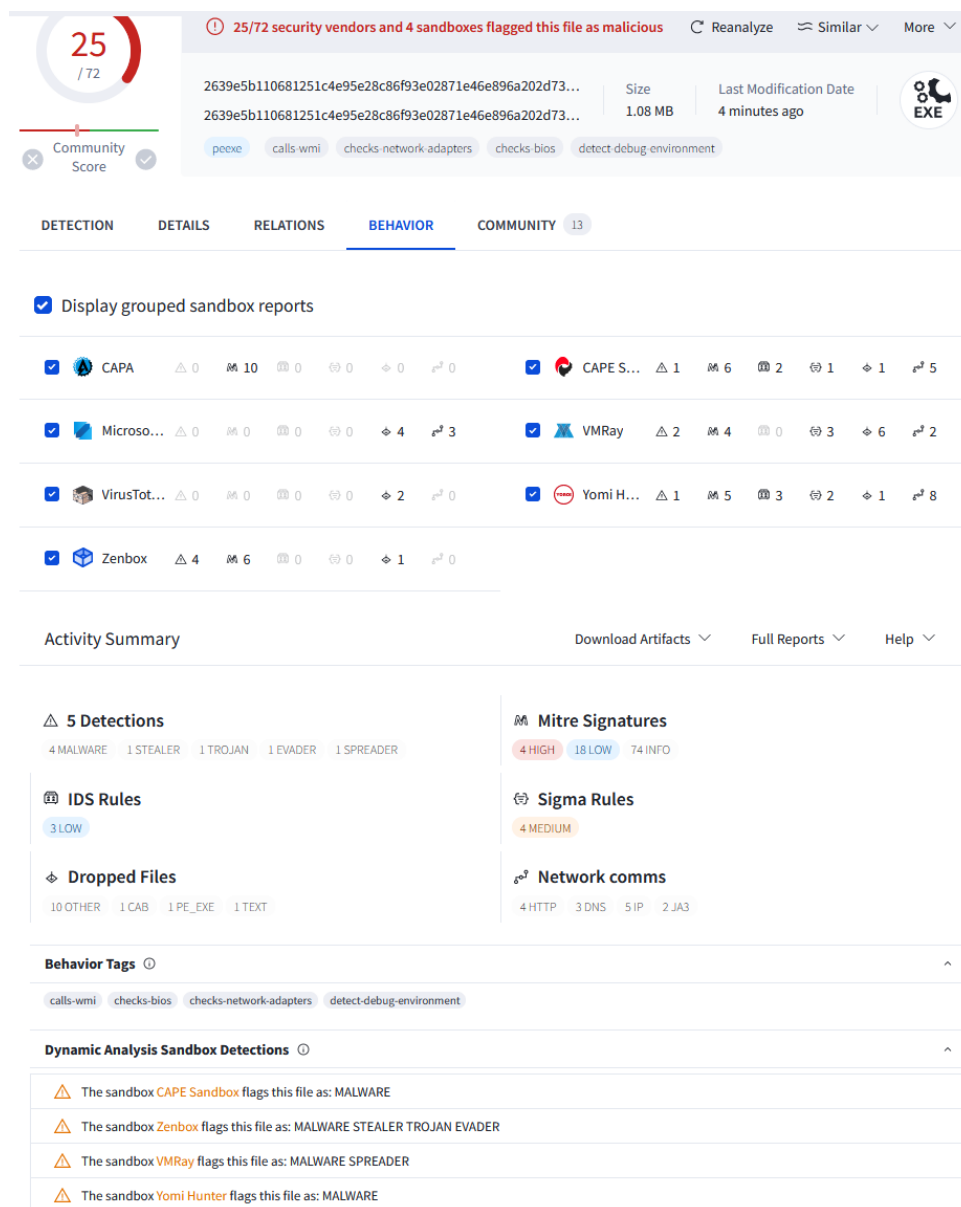
Continuing down in the relations tab in Figure 14, we see some more benign-looking results. However, pay very close attention and take note of the “Dropped Files” section. Upon first analysis, I overlooked this detail thinking it may not be too relevant. But when we dive more deeply into Basic Dynamic analysis, we will see how this becomes increasingly relevant.

Figure 14.



Under the behavior tab, we see additional information about what the sandbox environments found and concluded. As we can see, the dynamic analysis sandboxes in Figure 15 found that this executable is a MALWARE, and/or MALWARE STEALER TROJAN EVADER, and/or MALWARE SPREADER. Again, solidifying our assumption that this is a newer strain of malware that is difficult to detect using more methodical analysis employed by the undetected vendors.

Figure 15.



Finally, arriving at the community tab in Figure 16 shows us that while this strain of malware may be fairly new and undetected, its origin strain is very well known and documented in the “Contained in Collections” section. The middle link by JaffaCakes118 contains some interesting information as well which we will quickly look at next.

Figure 16.

2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2

25
/ 72

Community Score

25/72 security vendors and 4 sandboxes flagged this file as malicious

Reanalyze

Similar

More

2639e5b110681251c4e95e28c86f93e02871e46e896a202d73...

Size
1.08 MB

Last Modification Date
4 minutes ago

EXE

peexe

calls wmi

checks network adapters

checks bios

detect debug environment

DETECTION

DETAILS

RELATIONS

BEHAVIOR

COMMUNITY 13

Contained In Collections (3)

Agent Tesla

Updated 28 minutes ago by CarlosCabal

trusted A.NET based information stealer readily available to actors due to leaked builders. The malware is...

Files: 75.4 K URLs: 2.6 K Domains: 187 IPs: 732

Agent Tesla Malware

Updated 1 hour ago by JaffaCakes118

[b]A collection of files, url's, ip's and domains related to the malware Agent Tesla.[/b] [b]If you want to requ...

Files: 80.3 K URLs: 449 Domains: 17

AgentTesla

Updated 4 hours ago by NIXLovesXerneas

Agent Tesla is a remote access trojan (RAT) . It can collect data such as keystrokes and login credentials use...

Files: 3.3 K URLs: 12 Domains: 17 IPs: 3

Voting details (3)

NIXLovesXerneas

4 hours ago

-1

ker88

5 hours ago

-1

JaffaCakes118

5 hours ago

-9

Comments (7)

VMRay

2 hours ago

VMRay Analysis Verdict: Malicious

Threat Names: AgentTesla.v4, AgentTesla

Classifications: Injector, Spyware

Analysis Report: https://www.vmrays.com/analyses/_vt/2639e5b11068/report/overview.html

IOC Tab: https://www.vmrays.com/analyses/_vt/2639e5b11068/report/ioc.html

Function Log: https://www.vmrays.com/analyses/_vt/2639e5b11068/logs/flog.txt

STIX 2.0 IOCs: https://www.vmrays.com/analyses/_vt/2639e5b11068/report/artifacts/stix-report-2-0-iocs.json

summary.json: https://www.vmrays.com/analyses/_vt/2639e5b11068/logs/summary_v2.json

Reading over some community information provided by JaffaCakes118 in Figure 17 we can see some related file hashes. And there are quite a lot, more than 80,000 to be precise. This shows us how this malware has been blown out of proportion.

Figure 17.

Collection

Agent Tesla Malware

By JaffaCakes118

Share & Visibility

Download

Open in Graph

SUMMARY

IOCS

GRAPH

COMMUNITY

Related file hashes

(50 / 80.3 K)

Export

Tools

	Detections	Size	First seen	Last seen
0000361F90000CECA0B05E840FCA7B7CC535A0E786720... Result.exe peexe assembly direct-cpu-clock-access spreader detect-debug-environment ...	45 / 59	666.50 KB	2019-02-24 04:38:25	2024-02-12 17:20:31
0000E8BFDD41E01FB3D36315502AC76C5553905A3ABDD... eUZN.exe peexe assembly runtime-modules detect-debug-environment checks-network-adapters ...	57 / 72	1.10 MB	2023-03-29 08:46:44	2023-05-02 15:57:07
0003E2236ACD50EBD8D0944C7DC10BA37478D8E44EA4B... PnOv.exe peexe assembly detect-debug-environment checks-network-adapters checks-bios ...	50 / 72	1016.00 KB	2023-06-05 04:08:49	2023-06-05 23:24:19
000518A2AC056E4E5907AA1EAB5B5C4E61B728583A4F7... ...e5907aa1eab5b5c4e61b728583a4f77322a6c85190993f4df.exe peexe assembly invalid-signature overlay long-sleeps detect-debug-environment ...	48 / 69	2.43 MB	2023-07-13 15:24:08	2024-04-11 15:45:23
0005A5E48DA32AAA538FC76E9A742A1E8577368B2CD51... ...989e85c5ad1a917e9daf2d9e10c96af9e4f0e5449bfb527aa.zip zip encrypted	1 / 62	673.14 KB	2024-04-02 10:38:57	2024-04-02 10:38:57
0005EC4D705719B222998BC82EE69DD942E213C102F24... OrdinalComparer.exe peexe assembly runtime-modules detect-debug-environment checks-network-adapters ...	60 / 70	786.50 KB	2021-05-26 11:15:54	2022-11-21 08:18:59
0007F855C65B913A538D9A7680C5078DCDF8F49E26F37... ...a538d9a7680c5078dcdf8f49e26f37a130ec982c275bb693b.exe peexe overlay runtime-modules detect-debug-environment checks-network-adapters ...	56 / 71	276.80 KB	2023-03-03 09:31:32	2023-03-07 09:27:56
0008F04621E4F7495849729AEE51C17492B73B759B53E... YUP.exe peexe assembly runtime-modules detect-debug-environment checks-network-adapters ...	56 / 71	170.00 KB	2022-07-07 19:47:32	2022-11-19 14:30:04
0009426A08A10891A3268AF4B4E133BDC9AB5C5AE4DD8... Tup.exe peexe assembly runtime-modules detect-debug-environment spreader ...	53 / 69	939.00 KB	2022-02-14 09:19:24	2022-02-18 21:41:01

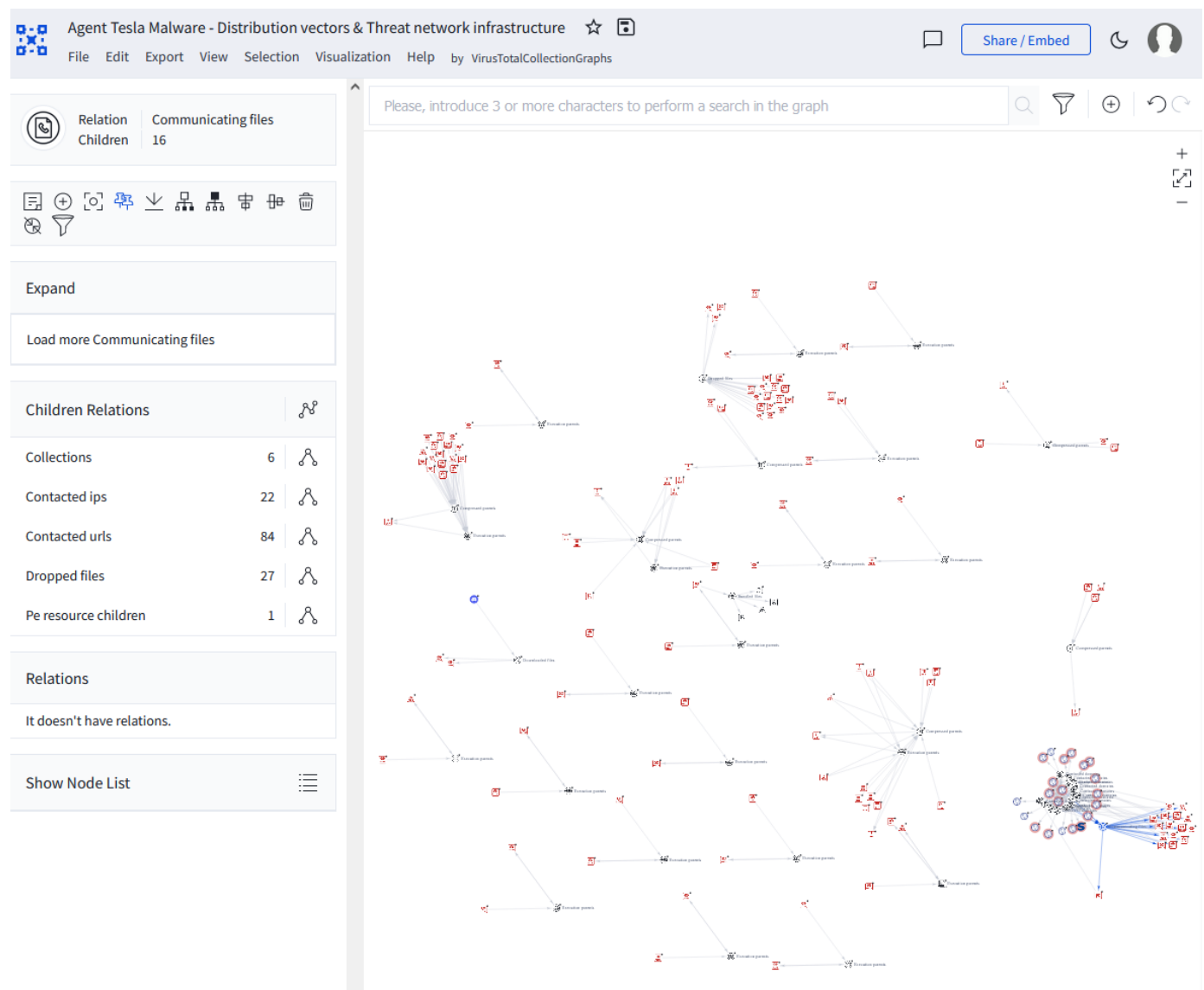
Additional information was provided as well in Figure 18, including Related domains that could be spreading the malware. I noticed Discord to be on the list as well which is interesting since Discord has been increasingly used by hackers to host “Webhooks” which are used almost like a Dropbox, where stolen user data gets sent to.

Figure 18.

Related domains (17 / 17)						Export ▾	Tools ▾
☐		Detections	Registrar	Created	Last Updated		
☐	aek0aicifaloh1yo.com 194.93.0.8 87.247.142.40 193.222.62.61 Malicious, Phishing (alphaMountain.ai) Phishing and Other Frauds spyware and malware dga	14 / 90	Atak Domain	2023-02-24 12:33:12	2024-03-16 00:34:19		
☐	alphastand.top 34.91.32.224 63.251.235.76 107.178.223.183 Bot Nets known infection source phishing and fraud	13 / 90	-	2023-06-20 00:00:00	2023-06-20 00:00:00		
☐	alphastand.trade 13.248.213.45 76.223.67.189 34.98.99.30 Malicious, Phishing (alphaMountain.ai) known infection source malware top-1M	16 / 90	GoDaddy.com, LLC	2020-03-23 23:54:45	2024-03-23 10:01:19		
☐	alphastand.win 107.178.223.183 104.155.138.21 35.204.181.10 Malicious, Suspicious (alphaMountain.ai) known infection source spyware and malware	16 / 90	Gandi SAS	2023-01-10 15:52:21	2024-01-30 11:53:35		
☐	api.telegram.org ↗ telegram.org 2001:67c:4e8:f004:c9 159.106.121.75 11.23.33.44 im online chat technical information top-100K	1 / 90	GoDaddy.com, LLC	2003-12-15 14:48:05	2023-07-17 23:35:03		
☐	discord.com 142.250.141.27 74.125.197.27 11.23.33.44 top-1K	0 / 90	CloudFlare, Inc.	2000-11-06 11:45:38	2021-09-28 14:47:55		
☐	discordapp.com 162.125.32.15 199.59.149.206 69.197.153.180 im online chat software-hardware top-1K	0 / 90	CloudFlare, Inc.	2015-02-26 22:47:43	2023-12-04 18:55:32		
☐	hie7doodohpae4na.com 81.94.159.129 194.93.0.8 45.91.8.120 Malicious, Phishing (alphaMountain.ai) Phishing and Other Frauds spyware and malware dga	14 / 90	Atak Domain	2023-02-24 12:33:20	2024-03-16 00:34:24		
☐	hoh0aeghwugh2gie.com 81.94.159.129 194.93.0.8 185.12.126.197 Malicious (alphaMountain.ai) known infection source spyware and malware dga	15 / 90	Atak Domain	2023-02-24 12:33:18	2024-03-16 00:34:23		
☐	host-file-host6.com 94.156.8.42 172.67.172.189 104.21.30.102 Malicious, Phishing (alphaMountain.ai) known infection source parked top-1M	16 / 90	-	2023-12-29 00:00:00	2023-12-30 00:00:00		

Finally, our last tab in Figure 19 contains a visual graph mapping of the malware. It provides us with a nice top-down view of how this malware is behaving. And again, as we see, it has many isolated instances of distribution vectors that correspond with our assumptions.

Figure 19.



AgentTesla: Reghot

Regshot allowed us to gain some insight as to how the malware operates, keep note that I ran Regshot twice to confirm my suspicions while also comparing some values to my host PC. It was the first tool that started allowing me to connect some dots and gain information about the logical behavior of the malware. While sifting through some of the entries in Figure 20 I noticed a reference to SMB1. Usually, SMB1 is disabled in modern Operating Systems by SMB2 and SMB3 since it is very vulnerable. However here we see that SMB1 usage is being tracked, and modified moments after the execution of the executable. Possibly indicating that there is an exploit being used to spread the malware throughout networks. The usage (counter) in our case has a hex value of E1 (225) jumping to a value of E2 (226) as highlighted below.

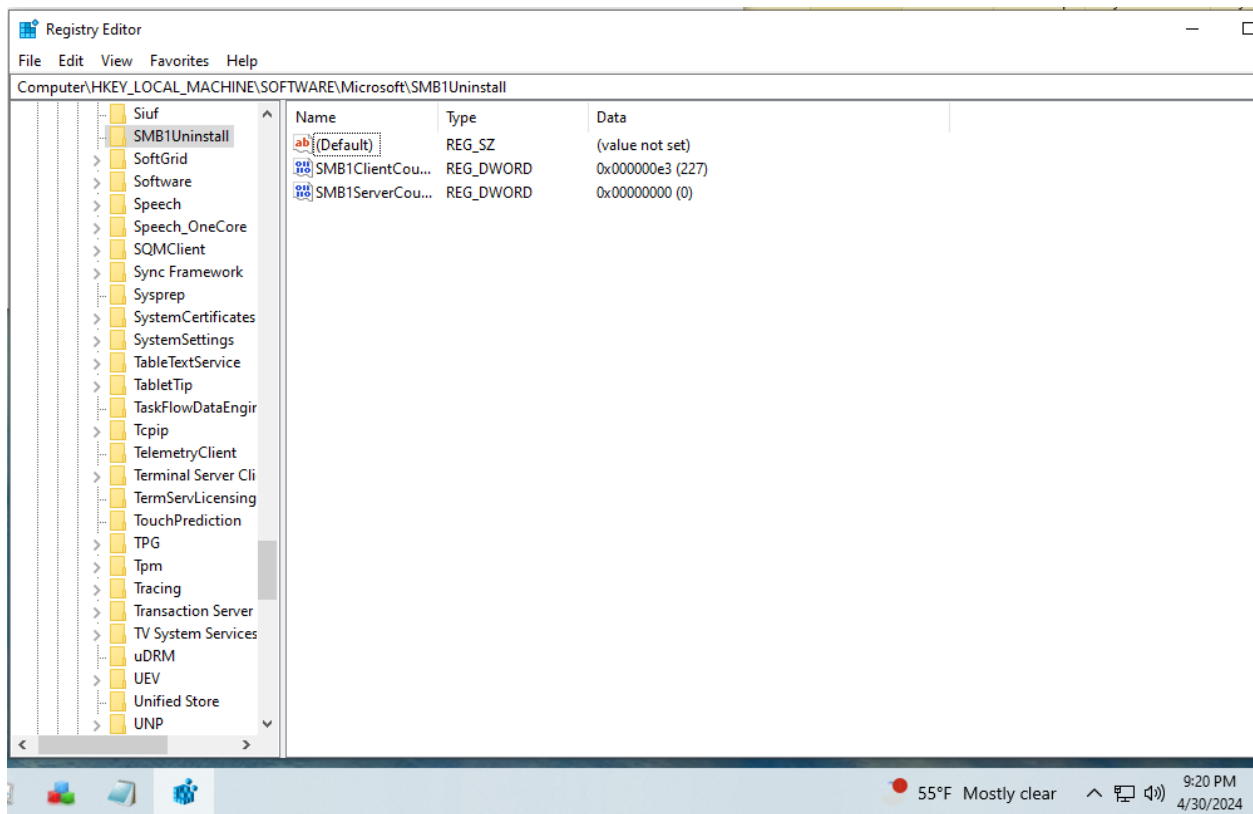
Figure 20.

[illegible]

Interestingly enough, a few minutes after the Regshot was done. I decided to observe the registry value in Registry Editor, that's where I noticed the value increment in Figure 21.

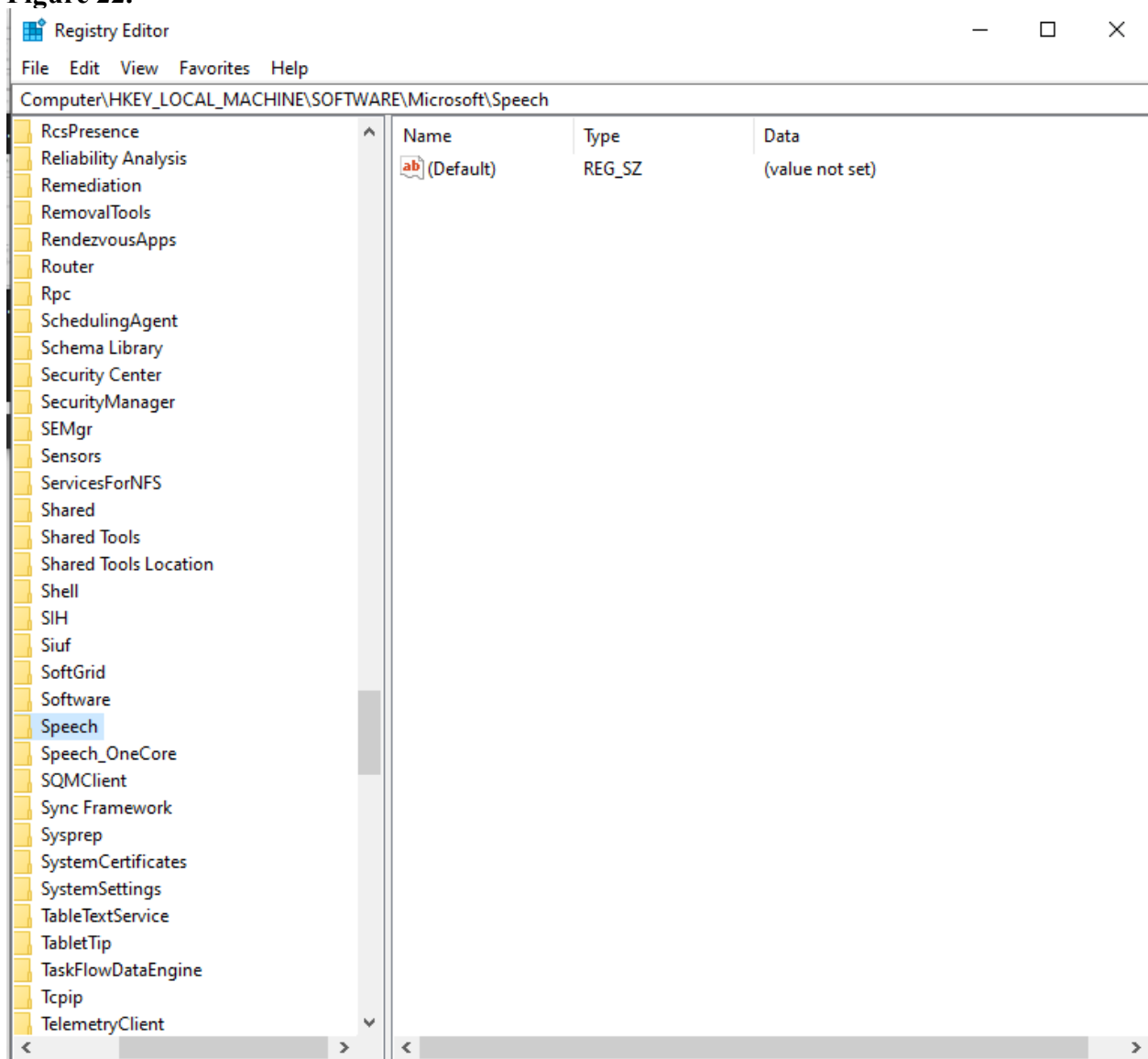
Indicating that a process is repeatedly trying to access SMB1 and incrementing it to E3 (227).

Figure 21.



To investigate the SMB1 registry entry, I compared it and its values between my host PC and the virtual machine. This comparison was important because all of the programs installed on the virtual machine (except the malware) were also installed on my host PC. Thus, any registry values present on the virtual machine but absent from the host PC would suggest that these were introduced by the malware, as it is the only differing factor. And as we can see in Figure 22, this is exactly the case. The host PC does not even contain the path for SMB1, indicating that the malware is accessing this outdated vulnerable protocol.

Figure 22.



AgentTesla: Process Monitor

Process Monitor provided a goldmine of information. I did my best to omit and condense as much as I could. However, upon execution of the malware, a long list of operations was executed. The reason why I won't omit them is because I am certain they all compile and write their information in two separate files which also appear in the list of operations. The next Figures will illustrate this. The configuration for the filters will be provided at the end as well.

Figure 23.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Na...	PID	Path	Operation	Sequence		
C= 2639e5b11...	3596		Process Start	0		
C= 2639e5b11...	3596		Thread Create	1		
C= 2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	Load Image	2		
C= 2639e5b11...	3596	C:\Windows\System32\ntdll.dll	Load Image	3		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	Load Image	4		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	5		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	6		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossibleDeadlock	RegQueryValue	7		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	8		
C= 2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	9		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	10		
C= 2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	11		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	12		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	RegQueryValue	13		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	14		
C= 2639e5b11...	3596	C:\Windows	CreateFile	15		
C= 2639e5b11...	3596	C:\Windows\System32\wow64.dll	Load Image	16		
C= 2639e5b11...	3596	C:\Windows\System32\wow64win.dll	Load Image	17		
C= 2639e5b11...	3596	C:\Windows\System32\wow64log.dll	CreateFile	18		
C= 2639e5b11...	3596	C:\Windows	CreateFile	19		
C= 2639e5b11...	3596	C:\Windows	QueryNameInformationFile	20		
C= 2639e5b11...	3596	C:\Windows	CloseFile	21		
C= 2639e5b11...	3596	HKLM\Software\Microsoft\Wow64\x86	RegOpenKey	22		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Wow64\x86\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	RegQueryValue	23		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Wow64\x86(Default)	RegQueryValue	24		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Wow64\x86	RegCloseKey	25		
C= 2639e5b11...	3596	C:\Windows\System32\wow64cpu.dll	Load Image	26		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	27		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	28		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	29		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossibleDeadlock	RegQueryValue	30		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	31		
C= 2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	32		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	33		
C= 2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	34		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	35		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	36		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	RegQueryValue	37		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	38		
C= 2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)	CreateFile	39		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\kernel32.dll	Load Image	40		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\KernelBase.dll	Load Image	41		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\StateSeparation\RedirectionMap\Keys	RegOpenKey	42		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\StateSeparation\RedirectionMap\Keys	RegOpenKey	43		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	44		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	45		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	RegOpenKey	46		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	RegOpenKey	47		
C= 2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	48		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	49		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegSetInfoKey	50		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\TransparentEnabled	RegQueryValue	51		
C= 2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegCloseKey	52		
C= 2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	53		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\FileSystem\	RegOpenKey	54		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\FileSystem	RegOpenKey	55		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\FileSystem	RegSetInfoKey	56		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\FileSystem\LongPathsEnabled	RegQueryValue	57		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\FileSystem	RegCloseKey	58		
C= 2639e5b11...	3596		Thread Create	59		
C= 2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide\AssemblyStorageRoots	RegOpenKey	60		
C= 2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b6144ccf1df_6.0.19041.3636_none_a8...	CreateFile	61		
C= 2639e5b11...	3596		Thread Create	62		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	63		
C= 2639e5b11...	3596		Thread Create	64		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\psapi.dll	Load Image	65		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	66		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	67		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\SafeDllSearchMode	RegQueryValue	68		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\user32.dll	Load Image	69		
C= 2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\VERSION.dll	CreateFile	70		
C= 2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CreateFile	71		
C= 2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	72		
C= 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	73		
Showing 825 of 764,675 events (0.10%)					Backed by virtual memory	

Figure 24.

Process Monitor - Sysinternals: www.sysinternals.com					
File Edit Event Filter Tools Options Help					
Process Name	PID	Path	Operation	Sequence	
C:\2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	72	
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	73	
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	74	
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	RegQueryValue	75	
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	76	
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\WINMM.dll	CreateFile	77	
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\WSOCK32.dll	CreateFile	78	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	QueryBasicInformationFile	79	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CloseFile	80	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\win32u.dll	Load Image	81	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\gdi32.dll	Load Image	82	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\gdi32full.dll	Load Image	83	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CreateFile	84	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CreateFileMapping	85	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CreateFileMapping	86	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\msvc_p_win.dll	Load Image	87	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	Load Image	88	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ucrtbase.dll	Load Image	89	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\msvcrt.dll	Load Image	90	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CreateFile	91	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\version.dll	CloseFile	92	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	QueryBasicInformationFile	93	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CloseFile	94	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CreateFile	95	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	QueryBasicInformationFile	96	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CloseFile	97	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CreateFile	98	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CreateFileMapping	99	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CreateFileMapping	100	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	Load Image	101	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\comdlg32.dll	Load Image	102	
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CloseFile	103	
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\MPR.dll	CreateFile	104	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\combase.dll	Load Image	105	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\vpert4.dll	Load Image	106	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\SHCore.dll	Load Image	107	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CreateFile	108	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	QueryBasicInformationFile	109	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CloseFile	110	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\shlwapi.dll	Load Image	111	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CreateFile	112	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CreateFileMapping	113	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CreateFileMapping	114	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\shell32.dll	Load Image	115	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	Load Image	116	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\winmm.dll	CloseFile	117	
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\WININET.dll	CreateFile	118	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\advapi32.dll	Load Image	119	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\sechost.dll	Load Image	120	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\bcrypt.dll	Load Image	121	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CreateFile	122	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CreateFile	123	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	QueryBasicInformationFile	124	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CloseFile	125	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CreateFileMapping	126	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CreateFileMapping	127	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CreateFile	128	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ole32.dll	Load Image	129	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	Load Image	130	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	CreateFile	131	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CreateFileMapping	132	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CreateFileMapping	133	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	QueryBasicInformationFile	134	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	CloseFile	135	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\oleaut32.dll	Load Image	136	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	Load Image	137	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ws2_32.dll	Load Image	138	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	CreateFile	139	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\mpr.dll	CloseFile	140	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	CreateFileMapping	141	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	CreateFileMapping	142	
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\IPHLPAPI.DLL	CreateFile	143	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wsck32.dll	CloseFile	144	
C:\2639e5b11...	3596	C:\Windows\SysWOW64\wininet.dll	Load Image	145	
Showing 825 of 812,753 events (0.10%)				Backed by virtual memory	

Figure 25.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Name	PID	Path	Operation	Sequence		
C:\Windows\SysWOW64\wsock32.dll	3596	C:\Windows\SysWOW64\wsock32.dll	CloseFile	144		
C:\Windows\SysWOW64\wininet.dll	3596	C:\Windows\SysWOW64\wininet.dll	Load Image	145		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CreateFile	146		
C:\Users\vbouser\Desktop\New folder (2)\USERENV.dll	3596	C:\Users\vbouser\Desktop\New folder (2)\USERENV.dll	CreateFile	147		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	QueryBasicInformationFile	148		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CloseFile	149		
C:\Users\vbouser\Desktop\New folder (2)\UxTheme.dll	3596	C:\Users\vbouser\Desktop\New folder (2)\UxTheme.dll	CreateFile	150		
C:\Windows\SysWOW64\wininet.dll	3596	C:\Windows\SysWOW64\wininet.dll	CloseFile	151		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CreateFile	152		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CreateFileMapping	153		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CreateFileMapping	154		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	Load Image	155		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CreateFile	156		
C:\Windows\SysWOW64\IPHLPAPI.DLL	3596	C:\Windows\SysWOW64\IPHLPAPI.DLL	CloseFile	157		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	QueryBasicInformationFile	158		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CloseFile	159		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CreateFile	160		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	QueryBasicInformationFile	161		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CloseFile	162		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CreateFile	163		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CreateFileMapping	164		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CreateFileMapping	165		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CreateFile	166		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CreateFileMapping	167		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	Load Image	168		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CreateFileMapping	169		
C:\Windows\SysWOW64\userenv.dll	3596	C:\Windows\SysWOW64\userenv.dll	CloseFile	170		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	Load Image	171		
C:\Windows\SysWOW64\uxtheme.dll	3596	C:\Windows\SysWOW64\uxtheme.dll	CloseFile	172		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegOpenKey	173		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegOpenKey	174		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegSetInfoKey	175		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegQueryValue	176		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\DXINXOptions	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\DXINXOptions	RegOpenKey	177		
HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	RegOpenKey	178		
HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	RegOpenKey	179		
HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	RegSetInfoKey	180		
HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\Default	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\Default	RegQueryValue	181		
HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\000603xx	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\000603xx	RegQueryValue	182		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CreateFile	183		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	QueryBasicInformationFile	184		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CloseFile	185		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CreateFile	186		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CreateFileMapping	187		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	QueryStandardInformationFile	188		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CreateFileMapping	189		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	CloseFile	190		
C:\Windows\SysWOW64\imm32.dll	3596	C:\Windows\SysWOW64\imm32.dll	Load Image	191		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	RegOpenKey	192		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegOpenKey	193		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	RegQueryValue	194		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	RegOpenKey	195		
HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	RegOpenKey	196		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	RegOpenKey	197		
HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	RegOpenKey	198		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	RegOpenKey	199		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegQueryValue	200		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2.exe	RegOpenKey	201		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	RegOpenKey	202		
HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	RegOpenKey	203		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Display	RegOpenKey	204		
HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Display	RegOpenKey	205		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	206		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	207		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegSetInfoKey	208		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegQueryValue	209		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegCloseKey	210		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	211		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	212		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegSetInfoKey	213		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegQueryValue	214		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegCloseKey	215		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	216		
HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	217		

Showing 825 of 844,398 events (0.097%)

Backed by virtual memory

Figure 26.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Na...	PID	Path	Operation	Sequence		
2639e5b11...	3596	HKLM\Software\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b110681251c4...	RegOpenKey	216		
2639e5b11...	3596	HKLM\Software\Policies\Microsoft\Windows\Control Panel\Desktop	RegOpenKey	217		
2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop	RegOpenKey	218		
2639e5b11...	3596	HKCU\Control Panel\Desktop	RegOpenKey	219		
2639e5b11...	3596	HKCU\Control Panel\Desktop\EnablePerProcessSystemDPI	RegQueryValue	220		
2639e5b11...	3596	HKCU\Control Panel\Desktop	RegCloseKey	221		
2639e5b11...	3596	HKLM\Software\Wow6432Node\Microsoft\Windows NT\CurrentVersion\Compatibility32	RegOpenKey	222		
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Compatibility32\2639e5b11068...	RegQueryValue	223		
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Compatibility32	RegCloseKey	224		
2639e5b11...	3596	HKLM\Software\Wow6432Node\Microsoft\Windows NT\CurrentVersion\IME Compatibility	RegOpenKey	225		
2639e5b11...	3596	C:\Windows\System32\wow64win.dll	ReadFile	226		
2639e5b11...	3596	C:\Windows\System32\wow64win.dll	ReadFile	227		
2639e5b11...	3596	C:\Windows\System32\wow64win.dll	ReadFile	228		
2639e5b11...	3596	C:\Windows\System32\wow64win.dll	ReadFile	229		
2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Nls\CustomLocale	RegOpenKey	230		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegOpenKey	231		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegSetInfoKey	232		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale\EMPTY	RegQueryValue	233		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale\EMPTY	RegQueryValue	234		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\NLS\Language	RegOpenKey	235		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\NLS\Language	RegOpenKey	236		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\NLS\Language	RegOpenKey	237		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Language\InstallLanguageFallback	RegQueryValue	238		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Language	RegCloseKey	239		
2639e5b11...	3596	HKLM\OSDATA\System\CurrentControlSet\Control\MUI\UILanguages	RegOpenKey	240		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegOpenKey	241		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegOpenKey	242		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegSetInfoKey	243		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegEnumKey	244		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegQueryKey	245		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\en-US	RegOpenKey	246		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\en-US>Type	RegQueryValue	247		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\en-US\AlternateCodePage	RegQueryValue	248		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\en-US	RegCloseKey	249		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegEnumKey	250		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages	RegCloseKey	251		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\PendingDelete	RegOpenKey	252		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\UILanguages\PendingDelete	RegOpenKey	253		
2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\MUI\Settings	RegOpenKey	254		
2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\MUI\Settings	RegOpenKey	255		
2639e5b11...	3596	HKCU	RegOpenKey	256		
2639e5b11...	3596	HKCU	RegQueryKey	257		
2639e5b11...	3596	HKCU	RegQueryKey	258		
2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached\MachineLanguageConfiguration	RegOpenKey	259		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\Settings\LanguageConfiguration	RegOpenKey	260		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\Settings\LanguageConfiguration	RegOpenKey	261		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\Settings\LanguageConfiguration	RegSetInfoKey	262		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\Settings\LanguageConfiguration	RegEnumValue	263		
2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\MUI\Settings\LanguageConfiguration	RegCloseKey	264		
2639e5b11...	3596	HKCU	RegCloseKey	265		
2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\MUI\Settings	RegOpenKey	266		
2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\MUI\Settings	RegOpenKey	267		
2639e5b11...	3596	HKCU	RegOpenKey	268		
2639e5b11...	3596	HKCU	RegQueryKey	269		
2639e5b11...	3596	HKCU	RegQueryKey	270		
2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Control Panel\Desktop	RegOpenKey	271		
2639e5b11...	3596	HKCU	RegQueryKey	272		
2639e5b11...	3596	HKCU	RegQueryKey	273		
2639e5b11...	3596	HKCU\Control Panel\Desktop\LanguageConfiguration	RegOpenKey	274		
2639e5b11...	3596	HKCU	RegCloseKey	275		
2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\MUI\Settings	RegOpenKey	276		
2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\MUI\Settings	RegOpenKey	277		
2639e5b11...	3596	HKCU	RegOpenKey	278		
2639e5b11...	3596	HKCU	RegQueryKey	279		
2639e5b11...	3596	HKCU	RegQueryKey	280		
2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Control Panel\Desktop	RegOpenKey	281		
2639e5b11...	3596	HKCU	RegQueryKey	282		
2639e5b11...	3596	HKCU	RegQueryKey	283		
2639e5b11...	3596	HKCU\Control Panel\Desktop	RegOpenKey	284		
2639e5b11...	3596	HKCU\Control Panel\Desktop	RegSetInfoKey	285		
2639e5b11...	3596	HKCU\Control Panel\Desktop\PreferredUILanguages	RegQueryValue	286		
2639e5b11...	3596	HKCU\Control Panel\Desktop	RegCloseKey	287		
2639e5b11...	3596	HKCU	RegCloseKey	288		
2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\MUI\Settings	RegOpenKey	289		
Showing 825 of 881,591 events (0.093%)					Backed by virtual memory	

Figure 27.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Na...	PID	Path	Operation	Sequence		
C:\2639e5b11...	3596	HKCU	RegCloseKey	288		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\MUI\Settings	RegOpenKey	289		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Policies\Microsoft\MUI\Settings	RegOpenKey	290		
C:\2639e5b11...	3596	HKCU	RegOpenKey	291		
C:\2639e5b11...	3596	HKCU	RegQueryKey	292		
C:\2639e5b11...	3596	HKCU	RegQueryKey	293		
C:\2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached	RegOpenKey	294		
C:\2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached	RegSetInfoKey	295		
C:\2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached\MachinePreferredUILanguages	RegQueryValue	296		
C:\2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached\MachinePreferredUILanguages	RegQueryValue	297		
C:\2639e5b11...	3596	HKCU\Control Panel\Desktop\MuiCached	RegCloseKey	298		
C:\2639e5b11...	3596	HKCU	RegCloseKey	299		
C:\2639e5b11...	3596	HKLM	RegOpenKey	300		
C:\2639e5b11...	3596	HKLM	RegQueryKey	301		
C:\2639e5b11...	3596	HKLM	RegQueryKey	302		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Windows	RegOpenKey	303		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Windows	RegSetInfoKey	304		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Windows\LoadApplint_DLLs	RegQueryValue	305		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows NT\CurrentVersion\Windows	RegCloseKey	306		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegQueryKey	307		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\2639e5b11068125...	RegOpenKey	308		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	CreateFile	309		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	CreateFileMapping	310		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	QueryStandardInformatio...	311		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	CreateFileMapping	312		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegOpenKey	313		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegSetInfoKey	314		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide\PreferExternalManifest	RegQueryValue	315		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegCloseKey	316		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	QueryStandardInformatio...	317		
C:\2639e5b11...	3596	C:\Windows\WindowsShell.Manifest	CloseFile	318		
C:\2639e5b11...	3596	HKLM	RegQueryKey	319		
C:\2639e5b11...	3596	HKLM	RegQueryKey	320		
C:\2639e5b11...	3596	HKLM\system\CurrentControlSet\control\Network Provider\HwOrder	RegOpenKey	321		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\control\Network Provider\HwOrder	RegOpenKey	322		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Network Provider\HwOrder	RegSetInfoKey	323		
C:\2639e5b11...	3596	HKLM	RegQueryKey	324		
C:\2639e5b11...	3596	HKLM	RegQueryKey	325		
C:\2639e5b11...	3596	HKLM\system\CurrentControlSet\control\Network Provider\ProviderOrder	RegOpenKey	326		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\control\Network Provider\ProviderOrder	RegOpenKey	327		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Network Provider\ProviderOrder	RegSetInfoKey	328		
C:\2639e5b11...	3596	HKLM	RegQueryKey	329		
C:\2639e5b11...	3596	HKLM	RegQueryKey	330		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\OLE	RegOpenKey	331		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegOpenKey	332		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegSetInfoKey	333		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole\PageAllocatorUseSystemHeap	RegQueryValue	334		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegCloseKey	335		
C:\2639e5b11...	3596	HKLM	RegQueryKey	336		
C:\2639e5b11...	3596	HKLM	RegQueryKey	337		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\OLE	RegOpenKey	338		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegOpenKey	339		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegSetInfoKey	340		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole\PageAllocatorSystemHeapsPrivate	RegQueryValue	341		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegCloseKey	342		
C:\2639e5b11...	3596	HKLM	RegQueryKey	343		
C:\2639e5b11...	3596	HKLM	RegQueryKey	344		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\OLE	RegOpenKey	345		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegOpenKey	346		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegSetInfoKey	347		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole\AggressiveMTATesting	RegQueryValue	348		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegCloseKey	349		
C:\2639e5b11...	3596	HKLM	RegQueryKey	350		
C:\2639e5b11...	3596	HKLM	RegOpenKey	351		
C:\2639e5b11...	3596	HKLM	RegSetInfoKey	352		
C:\2639e5b11...	3596	HKLM	RegQueryKey	353		
C:\2639e5b11...	3596	HKLM\Software\Microsoft\Ole\FeatureDevelopmentProperties	RegOpenKey	354		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\AppModel\Lookaside\Packages	RegOpenKey	355		
C:\2639e5b11...	3596	HKLM	RegQueryKey	356		
C:\2639e5b11...	3596	HKLM\Software\Microsoft\Ole\FeatureDevelopmentProperties	RegOpenKey	357		
C:\2639e5b11...	3596	HKLM	RegQueryKey	358		
C:\2639e5b11...	3596	HKLM\Software\Microsoft\Ole	RegOpenKey	359		
C:\2639e5b11...	3596	HKCU	RegOpenKey	360		
C:\2639e5b11...	3596	HKCU	RegSetInfoKey	361		
C:\2639e5b11...	3596	HKCU	RegQueryKey	362		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegOpenKey	363		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegOpenKey	364		
Showing 825 of 1,378,492 events (0.059%)					Backed by virtual memory	

Figure 28.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Na...	PID	Path	Operation	Sequence		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegOpenKey	364		
C:\2639e5b11...	3596	HKCU	RegCloseKey	365		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	366		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	367		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft\Ole\FeatureDevelopmentProperties	RegOpenKey	368		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	369		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	370		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft\Ole\FeatureDevelopmentProperties	RegOpenKey	371		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	372		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	373		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft\Ole	RegOpenKey	374		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	375		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegQueryKey	376		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft	RegOpenKey	377		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft	RegSetInfoKey	378		
C:\2639e5b11...	3596	HKLM	RegQueryKey	379		
C:\2639e5b11...	3596	HKLM	RegQueryKey	380		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\OLE\Tracing	RegOpenKey	381		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole\Tracing	RegOpenKey	382		
C:\2639e5b11...	3596	HKLM	RegQueryKey	383		
C:\2639e5b11...	3596	HKLM	RegQueryKey	384		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\OLE\Tracing	RegOpenKey	385		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole\Tracing	RegOpenKey	386		
C:\2639e5b11...	3596	HKLM	RegQueryKey	387		
C:\2639e5b11...	3596	HKLM	RegQueryKey	388		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\OLEAUT	RegOpenKey	389		
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c8f93e02871e46e896a202d731c...	QueryNameInformationFile	390		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\vpccs.dll	CreateFile	391		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CreateFile	392		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	QueryBasicInformationFile	393		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CloseFile	394		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CreateFile	395		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CreateFileMapping	396		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CreateFileMapping	397		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	Load Image	398		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\kernel.appcore.dll	CloseFile	399		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\bccrprimitives.dll	Load Image	400		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegOpenKey	401		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegOpenKey	402		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegSetInfoKey	403		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy\STE	RegQueryValue	404		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegCloseKey	405		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegOpenKey	406		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegOpenKey	407		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegSetInfoKey	408		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy\Enabled	RegQueryValue	409		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa	RegOpenKey	410		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa	RegOpenKey	411		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa	RegSetInfoKey	412		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegQueryValue	413		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy\MDMEnabled	RegQueryValue	414		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa\FipsAlgorithmPolicy	RegCloseKey	415		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Lsa	RegCloseKey	416		
C:\2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Policies\Microsoft\Cryptography\Configuration	RegOpenKey	417		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Policies\Microsoft\Cryptography\Configuration	RegOpenKey	418		
C:\2639e5b11...	3596	C:\Windows\System32\wow64win.dll	ReadFile	419		
C:\2639e5b11...	3596	HKCU	RegOpenKey	420		
C:\2639e5b11...	3596	HKCU	RegQueryKey	421		
C:\2639e5b11...	3596	HKCU	RegQueryKey	422		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize	RegOpenKey	423		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize	RegSetInfoKey	424		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize\AppsUseLightTheme	RegQueryValue	425		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Themes\Personalize	RegCloseKey	426		
C:\2639e5b11...	3596	HKCU	RegCloseKey	427		
C:\2639e5b11...	3596	HKCU	RegOpenKey	428		
C:\2639e5b11...	3596	HKCU	RegQueryKey	429		
C:\2639e5b11...	3596	HKCU	RegQueryKey	430		
C:\2639e5b11...	3596	HKCU\Control Panel\Mouse	RegOpenKey	431		
C:\2639e5b11...	3596	HKCU\Control Panel\Mouse	RegSetInfoKey	432		
C:\2639e5b11...	3596	HKCU\Control Panel\Mouse\SwapMouseButtons	RegQueryValue	433		
C:\2639e5b11...	3596	HKCU\Control Panel\Mouse	RegCloseKey	434		
C:\2639e5b11...	3596	HKCU	RegQueryKey	435		
C:\2639e5b11...	3596	HKCU	RegQueryKey	436		
C:\2639e5b11...	3596	HKCU\Software\AutoIt v3\AutoIt	RegOpenKey	437		
C:\2639e5b11...	3596	HKLM\Software\Policies\Microsoft\Windows\Control Panel\Desktop	RegOpenKey	438		
C:\2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop	RegOpenKey	439		
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c8f93e02871e46e896a202d731c...	CreateFile	440		

Showing 825 of 1,399,519 events (0.058%) Backed by virtual memory

Figure 29.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Name	PID	Path	Operation	Sequence		
C:\2639e5b11...	3596	HKCU\Software\Policies\Microsoft\Windows\Control Panel\Desktop	RegOpenKey	439		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	440		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	441		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryBasicInformationFile	442		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	443		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	444		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	445		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryBasicInformationFile	446		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	447		
C:\2639e5b11...	3596	C:\	CreateFile	448		
C:\2639e5b11...	3596	C:\Users	QueryDirectory	449		
C:\2639e5b11...	3596	C:\	CloseFile	450		
C:\2639e5b11...	3596	C:\Users	CreateFile	451		
C:\2639e5b11...	3596	C:\Users\vboxuser	QueryDirectory	452		
C:\2639e5b11...	3596	C:\Users	CloseFile	453		
C:\2639e5b11...	3596	C:\Users\vboxuser	CreateFile	454		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop	QueryDirectory	455		
C:\2639e5b11...	3596	C:\Users\vboxuser	CloseFile	456		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	457		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	458		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryBasicInformationFile	459		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	460		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	461		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\msctf.dll	Load Image	462		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegOpenKey	463		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegOpenKey	464		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegSetInfoKey	465		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale\en-US	RegQueryValue	466		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegCloseKey	467		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\ExtendedLocale	RegOpenKey	468		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\ExtendedLocale	RegOpenKey	469		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\ExtendedLocale	RegSetInfoKey	470		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\ExtendedLocale\en-US	RegQueryValue	471		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\ExtendedLocale	RegCloseKey	472		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions\000603xx	RegQueryValue	473		
C:\2639e5b11...	3596	C:\Windows\Globalization\Sorting\SortDefault.nls	CreateFile	474		
C:\2639e5b11...	3596	C:\Windows\Globalization\Sorting\SortDefault.nls	CreateFileMapping	475		
C:\2639e5b11...	3596	C:\Windows\Globalization\Sorting\SortDefault.nls	QueryStandardInformation...	476		
C:\2639e5b11...	3596	C:\Windows\Globalization\Sorting\SortDefault.nls	CreateFileMapping	477		
C:\2639e5b11...	3596	C:\Windows\Globalization\Sorting\SortDefault.nls	CloseFile	478		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\lds	RegOpenKey	479		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\lds	RegOpenKey	480		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\lds	RegSetInfoKey	481		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\lds\en-US	RegQueryValue	482		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\lds\en	RegQueryValue	483		
C:\2639e5b11...	3596	HKLM	RegQueryKey	484		
C:\2639e5b11...	3596	HKLM	RegQueryKey	485		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\CTF\Compatibility\2639e5b110681251c4e95e28c88f93e028...	RegOpenKey	486		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegOpenKey	487		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegSetInfoKey	488		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide\PreferExternalManifest	RegQueryValue	489		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegCloseKey	490		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\UxTheme.dll Config	CreateFile	491		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\uxtheme.dll	CreateFile	492		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\uxtheme.dll	CloseFile	493		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide\AssemblyStorageRoots	RegOpenKey	494		
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CreateFile	495		
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CloseFile	496		
C:\2639e5b11...	3596	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	497		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	498		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	499		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c393e647b08f2.exe	RegQueryValue	500		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	501		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryNameInformationFile	502		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryNameInformationFile	503		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CreateFile	504		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	QueryBasicInformationFile	505		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a202d731c...	CloseFile	506		
C:\2639e5b11...	3596	C:\	CreateFile	507		
C:\2639e5b11...	3596	C:\Users	QueryDirectory	508		
C:\2639e5b11...	3596	C:\	CloseFile	509		
C:\2639e5b11...	3596	C:\Users	CreateFile	510		
C:\2639e5b11...	3596	C:\Users\vboxuser	QueryDirectory	511		
C:\2639e5b11...	3596	C:\Users	CloseFile	512		
C:\2639e5b11...	3596	C:\Users\vboxuser	CreateFile	513		
C:\2639e5b11...	3596	C:\Users\vboxuser\Desktop	QueryDirectory	514		
C:\2639e5b11...	3596	C:\Users\vboxuser	CloseFile	515		
Showing 825 of 1,425,834 events (0.057%)					Backed by virtual memory	

Figure 30.

Process Monitor - Sysinternals: www.sysinternals.com				
File Edit Event Filter Tools Options Help				
Process Name	PID	Path	Operation	Sequence
2639e5b11...	3596	C:\Users\vbouser\Desktop	QueryDirectory	514
2639e5b11...	3596	C:\Users\vbouser	CloseFile	515
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CreateFile	516
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	QueryBasicInformationFile	517
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CloseFile	518
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CreateFile	519
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CreateFileMapping	520
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CreateFileMapping	521
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	Load Image	522
2639e5b11...	3596	C:\Windows\SysWOW64\windows.storage.dll	CloseFile	523
2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\Wldp.dll	CreateFile	524
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CreateFile	525
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	QueryBasicInformationFile	526
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CloseFile	527
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CreateFile	528
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CreateFileMapping	529
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CreateFileMapping	530
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	Load Image	531
2639e5b11...	3596	C:\Windows\SysWOW64\wldp.dll	CloseFile	532
2639e5b11...	3596	HKLM	RegQuery/Key	533
2639e5b11...	3596	HKLM	RegQuery/Key	534
2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegOpenKey	535
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegSetInfoKey	536
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegQuery/Key	537
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegOpenKey	538
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegCloseKey	539
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	540
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	541
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	542
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	543
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	544
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	545
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	546
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	547
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	548
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	549
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	550
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	551
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	552
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	553
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	554
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	555
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	556
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	557
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	558
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	559
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Value	560
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegQuery/Key	561
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegOpenKey	562
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegCloseKey	563
2639e5b11...	3596	HKLM	RegQuery/Key	564
2639e5b11...	3596	HKLM	RegQuery/Key	565
2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion	RegOpenKey	566
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\ProgramFilesDir	RegQuery/Value	567
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion	RegCloseKey	568
2639e5b11...	3596	HKLM	RegQuery/Key	569
2639e5b11...	3596	HKLM	RegQuery/Key	570
2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegOpenKey	571
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegSetInfoKey	572
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegQuery/Key	573
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegOpenKey	574
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegCloseKey	575
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	576
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	577
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	578
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	579
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	580
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	581
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	582
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	583
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	584
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	585
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	586
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	587
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	588
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	589
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQuery/Value	590
Showing 4,976 of 1,605,977 events (0.30%) Backed by virtual memory				

Figure 31.

Process Monitor - Sysinternals: www.sysinternals.com				
File Edit Event View Filter Tools Options Help				
Process Name	PID	Path	Operation	Sequence
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	590
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	591
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	592
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	593
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	594
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	595
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryValue	596
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegQueryKey	597
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegOpenKey	598
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegCloseKey	599
2639e5b11...	3596	HKLM	RegQueryKey	600
2639e5b11...	3596	HKLM	RegQueryKey	601
2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegOpenKey	602
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegSetInfoKey	603
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegQueryKey	604
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegOpenKey	605
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions	RegCloseKey	606
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	607
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	608
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	609
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	610
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	611
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	612
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	613
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	614
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	615
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	616
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	617
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	618
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	619
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	620
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	621
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	622
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	623
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	624
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	625
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	626
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryValue	627
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegQueryKey	628
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegOpenKey	629
2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegCloseKey	630
2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a20d2731c...	CreateFile	631
2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a20d2731c...	CreateFile	632
2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a20d2731c...	QueryBasicInformationFile	633
2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c88f93e02871e46e896a20d2731c...	CreateFile	634
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CreateFile	635
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	QueryBasicInformationFile	636
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CloseFile	637
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp	CreateFile	638
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp	QueryBasicInformationFile	639
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp	CloseFile	640
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CreateFile	641
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CloseFile	642
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CreateFile	643
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	WriteFile	644
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	WriteFile	645
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	WriteFile	646
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	WriteFile	647
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CloseFile	648
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CreateFile	649
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CreateFile	650
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	651
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	652
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	653
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	WriteFile	654
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	655
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	656
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	WriteFile	657
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	WriteFile	658
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	659
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	ReadFile	660
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	WriteFile	661
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CloseFile	662
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	WriteFile	663
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CloseFile	664
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CreateFile	665
2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	QueryAttributeTanFile	666

Showing 4,976 of 1,637,314 events (0.30%)

Backed by virtual memory

Figure 32.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Name	PID	Path	Operation	Sequence		
C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CreateFile	665		
C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	QueryAttribute TagFile	666		
C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	<Unknown>	667		
C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ACA.tmp	CloseFile	668		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CreateFile	669		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	SetBasicInformationFile	670		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CloseFile	671		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	CloseFile	672		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	CreateFile	673		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	CreateFile	674		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	QueryBasicInformationFile	675		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	CloseFile	676		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CreateFile	677		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	QueryBasicInformationFile	678		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CloseFile	679		
C:\Users\vbouser\AppData\Local\Temp	3596	C:\Users\vbouser\AppData\Local\Temp	CreateFile	680		
C:\Users\vbouser\AppData\Local\Temp	3596	C:\Users\vbouser\AppData\Local\Temp	QueryBasicInformationFile	681		
C:\Users\vbouser\AppData\Local\Temp	3596	C:\Users\vbouser\AppData\Local\Temp	CloseFile	682		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CreateFile	683		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CloseFile	684		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CreateFile	685		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	WriteFile	686		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	WriteFile	687		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CloseFile	688		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CreateFile	689		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CreateFile	690		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	ReadFile	691		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	ReadFile	692		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	ReadFile	693		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	WriteFile	694		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CloseFile	695		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	WriteFile	696		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CloseFile	697		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CreateFile	698		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	QueryAttribute TagFile	699		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	<Unknown>	700		
C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	3596	C:\Users\vbouser\AppData\Local\Temp\aut5ADA.tmp	CloseFile	701		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CreateFile	702		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	SetBasicInformationFile	703		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CloseFile	704		
C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	3596	C:\Users\vbouser\Desktop\New folder (2)\2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c...	CloseFile	705		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CreateFile	706		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	ReadFile	707		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	ReadFile	708		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	QueryStandardInformatio...	709		
C:\Users\vbouser\AppData\Local\Temp\camellin	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	ReadFile	710		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CreateFile	711		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	QueryStandardInformatio...	712		
C:\Users\vbouser\AppData\Local\Temp\myriopodous	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	ReadFile	713		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegQueryKey	714		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\RegSvc.exe	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\RegSvc.exe	RegOpenKey	715		
HKLM\Software\Microsoft\Wow64\vtajit	3596	HKLM\Software\Microsoft\Wow64\vtajit	RegOpenKey	716		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	CreateFile	717		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	CreateFileMapping	718		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	CreateFileMapping	719		
HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\RegSvc.exe	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options\RegSvc.exe	RegOpenKey	720		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	QuerySecurityFile	721		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	QueryNameInformationFile	722		
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	3596	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	RegOpenKey	723		
HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	3596	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	RegQueryValue	724		
HKLM\System\CurrentControlSet\Control\Session Manager\BAM	3596	HKLM\System\CurrentControlSet\Control\Session Manager\BAM	RegCloseKey	725		
HKLM\System\CurrentControlSet\Control\Session Manager\BAM	3596	HKLM\System\CurrentControlSet\Control\Session Manager\BAM	RegOpenKey	726		
HKLM\System\CurrentControlSet\Control\Session Manager\BAM	3596	HKLM\System\CurrentControlSet\Control\Session Manager\BAM	RegOpenKey	727		
C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvc.exe	Process Create	728		
HKLM\System\CurrentControlSet\Control\Session Manager\AppData	3596	HKLM\System\CurrentControlSet\Control\Session Manager\AppData	RegOpenKey	729		
HKLM\System\CurrentControlSet\Control\Session Manager\AppData	3596	HKLM\System\CurrentControlSet\Control\Session Manager\AppData	RegOpenKey	730		
HKLM\System\CurrentControlSet\Control\SafeBoot\Option	3596	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	731		
HKLM\System\CurrentControlSet\Control\SafeBoot\Option	3596	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	732		
HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Safer\CodeIdentifiers	3596	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	733		
HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	734		
HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegSetInfoKey	735		
HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\TransparentEnabled	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\TransparentEnabled	RegQueryValue	736		
HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\AuthenticodeEnabled	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\AuthenticodeEnabled	RegQueryValue	737		
HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	3596	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegCloseKey	738		
HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	3596	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegOpenKey	739		
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegOpenKey	740		
HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegSetInfoKey	741		
Showing 4,976 of 1,658,227 events (0.30%)					Backed by virtual memory	

Figure 33.

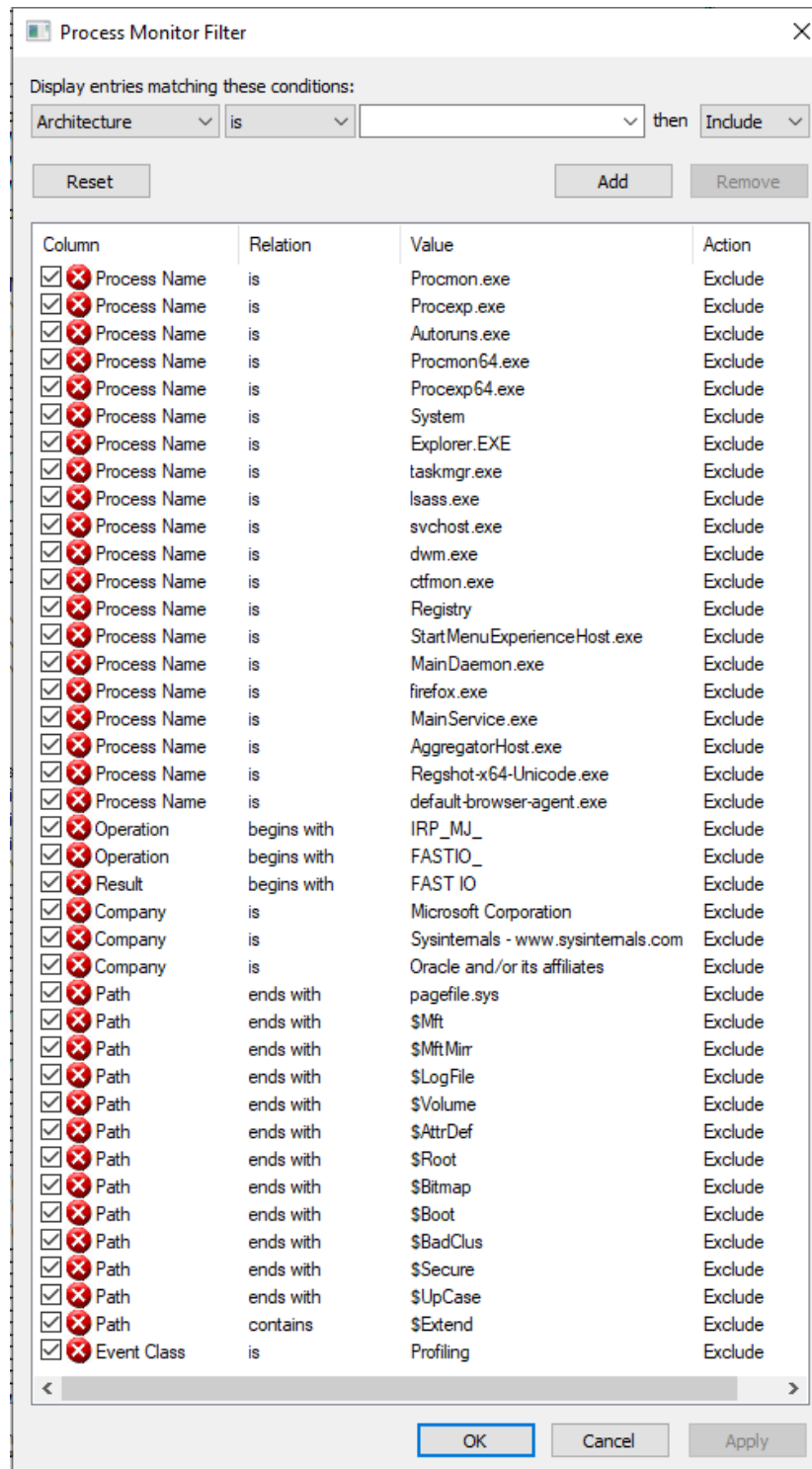
Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Name...	PID	Path	Operation	Sequence		
C:\2639e5b11...	3596	HKCU\Software\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegOpenKey	740		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegSetInfoKey	741		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders\Cache	RegQueryValue	742		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows\CurrentVersion\Explorer\Shell Folders	RegCloseKey	743		
C:\2639e5b11...	3596	HKCU\Software\Microsoft\Windows NT\CurrentVersion	RegOpenKey	744		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion	RegSetInfoKey	745		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion	RegQueryKey	746		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Layers	RegOpenKey	747		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Layers	RegSetInfoKey	748		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Layers\C:\Windows\Microsoft...	RegQueryValue	749		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion\AppCompatFlags\Layers	RegCloseKey	750		
C:\2639e5b11...	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe	QuerySecurityFile	751		
C:\2639e5b11...	3596	C:\Windows\apppatch\sysmain.sdb	CreateFile	752		
C:\2639e5b11...	3596	C:\Windows\apppatch\sysmain.sdb	QueryBasicInformationFile	753		
C:\2639e5b11...	3596	C:\Windows\apppatch\sysmain.sdb	CloseFile	754		
C:\2639e5b11...	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe	QueryBasicInformationFile	755		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegOpenKey	756		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegSetInfoKey	757		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide\PreferExternalManifest	RegQueryValue	758		
C:\2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\SideBySide	RegCloseKey	759		
C:\2639e5b11...	3596	C:\Windows\Microsoft.NET\Framework\v4.0.30319\RegSvcs.exe	CloseFile	760		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	761		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	762		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	763		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	764		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	765		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	766		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	767		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	768		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	769		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	770		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	771		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	772		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	773		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	774		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	775		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	776		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	777		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	778		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	779		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	780		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CreateFile	781		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	QueryStandardInformatio...	782		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	ReadFile	783		
C:\2639e5b11...	3596	C:\Windows\SysWOW64\ntdll.dll	CloseFile	784		
C:\2639e5b11...	3596		Thread Exit	785		
C:\2639e5b11...	3596		Thread Exit	786		
C:\2639e5b11...	3596		Thread Exit	787		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings\Software\Microsoft	RegCloseKey	788		
C:\2639e5b11...	3596	HKCU\Software\Classes\Local Settings	RegCloseKey	789		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Ole	RegCloseKey	790		
C:\2639e5b11...	3596	HKLM	RegCloseKey	791		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	792		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	793		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegSetInfoKey	794		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize\DisableMetaFiles	RegQueryValue	795		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegCloseKey	796		
C:\2639e5b11...	3596	HKLM\Software\WOW6432Node\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	797		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegOpenKey	798		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegSetInfoKey	799		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize\DisableUmpdBufferSizeCheck	RegQueryValue	800		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\GRE_Initialize	RegCloseKey	801		
C:\2639e5b11...	3596		Thread Exit	802		
C:\2639e5b11...	3596		Process Exit	803		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	RegOpenKey	804		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	RegQueryValue	805		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Services\bam\State\UserSettings\S-1-5-21-3187936749-114000109-41054...	RegCloseKey	806		
C:\2639e5b11...	3596	C:\Windows	CloseFile	807		
C:\2639e5b11...	3596	C:\Users\vbouser\Desktop\New folder (2)	CloseFile	808		
C:\2639e5b11...	3596	C:\Windows\WinSxS\x86_microsoft.windows.common-controls_6595b64144ccf1df_6.0.19041.3636_none_a8...	CloseFile	809		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	810		
C:\2639e5b11...	3596	HKLM\SOFTWARE\Microsoft\Windows NT\CurrentVersion\Image File Execution Options	RegCloseKey	811		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Versions	RegCloseKey	812		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\CustomLocale	RegCloseKey	813		
C:\2639e5b11...	3596	HKLM	RegCloseKey	814		
C:\2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion	RegCloseKey	815		
C:\2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Network Provider\HwOrder	RegCloseKey	816		
Showing 4,976 of 1,733,834 events (0.28%)					Backed by virtual memory	

Figure 34.

Process Monitor - Sysinternals: www.sysinternals.com						
File Edit Event Filter Tools Options Help						
Process Na...	PID	Path	Operation	Sequence		
C- 2639e5b11...	3596	HKCU\SOFTWARE\Microsoft\Windows NT\CurrentVersion	RegCloseKey	815		
C- 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Network Provider\HwOrder	RegCloseKey	816		
C- 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Network Provider\ProviderOrder	RegCloseKey	817		
C- 2639e5b11...	3596	HKCU	RegCloseKey	818		
C- 2639e5b11...	3596	HKLM\System\CurrentControlSet\Control\Nls\Sorting\Ids	RegCloseKey	819		
C- 2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\camellin	CloseFile	820		
C- 2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{D652...	RegCloseKey	821		
C- 2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{7C5A...	RegCloseKey	822		
C- 2639e5b11...	3596	HKLM\SOFTWARE\WOW6432Node\Microsoft\Windows\CurrentVersion\Explorer\FolderDescriptions\{F38B...	RegCloseKey	823		
C- 2639e5b11...	3596	C:\Users\vbouser\AppData\Local\Temp\myriopodous	CloseFile	824		
pingsender....	7976		Process Start	825		
pingsender....	7976		Thread Create	826		
pingsender....	7976	C:\Program Files (x86)\Mozilla Firefox\pingsender.exe	Load Image	827		
pingsender....	7976	C:\Windows\System32\ntdll.dll	Load Image	828		
pingsender....	7976	C:\Windows\SysWOW64\ntdll.dll	Load Image	829		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	830		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	831		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossibleDeadlock	RegQueryValue	832		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	833		
pingsender....	7976	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	834		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	835		
pingsender....	7976	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	836		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	837		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	RegQueryValue	838		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	839		
pingsender....	7976	C:\Windows	CreateFile	840		
pingsender....	7976	C:\Windows\System32\wow64.dll	Load Image	841		
pingsender....	7976	C:\Windows\System32\wow64win.dll	Load Image	842		
pingsender....	7976	C:\Windows\System32\wow64log.dll	CreateFile	843		
pingsender....	7976	C:\Windows	CreateFile	844		
pingsender....	7976	C:\Windows	QueryNameInformationFile	845		
pingsender....	7976	C:\Windows	CloseFile	846		
pingsender....	7976	HKLM\Software\Microsoft\Wow64\x86	RegOpenKey	847		
pingsender....	7976	HKLM\SOFTWARE\Microsoft\Wow64\x86\pingsender.exe	RegQueryValue	848		
pingsender....	7976	HKLM\SOFTWARE\Microsoft\Wow64\x86(Default)	RegQueryValue	849		
pingsender....	7976	HKLM\SOFTWARE\Microsoft\Wow64\x86	RegCloseKey	850		
pingsender....	7976	C:\Windows\System32\wow64cpu.dll	Load Image	851		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	852		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	853		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	854		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\RaiseExceptionOnPossibleDeadlock	RegQueryValue	855		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	856		
pingsender....	7976	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	857		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\Segment Heap	RegOpenKey	858		
pingsender....	7976	HKLM\SYSTEM\CurrentControlSet\Control\Session Manager	RegOpenKey	859		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	860		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	861		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\ResourcePolicies	RegQueryValue	862		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegCloseKey	863		
pingsender....	7976	C:\Windows\SysWOW64	CreateFile	864		
pingsender....	7976	C:\Windows\SysWOW64\kernel32.dll	Load Image	865		
pingsender....	7976	C:\Windows\SysWOW64\KernelBase.dll	Load Image	866		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\StateSeparation\RedirectionMap\Keys	RegOpenKey	867		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\StateSeparation\RedirectionMap\Keys	RegOpenKey	868		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	869		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\SafeBoot\Option	RegOpenKey	870		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	RegOpenKey	871		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Srp\GP\DLL	RegOpenKey	872		
pingsender....	7976	HKLM\Software\WOW6432Node\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	873		
pingsender....	7976	HKLM\SOFTWARE\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	874		
pingsender....	7976	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegSetInfoKey	875		
pingsender....	7976	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers\TransparentEnabled	RegQueryValue	876		
pingsender....	7976	HKLM\SOFTWARE\Policies\Microsoft\Windows\safer\codeidentifiers	RegCloseKey	877		
pingsender....	7976	HKCU\Software\Policies\Microsoft\Windows\Safer\CodeIdentifiers	RegOpenKey	878		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\FileSystem\	RegOpenKey	879		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\FileSystem	RegOpenKey	880		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\FileSystem	RegSetInfoKey	881		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\FileSystem\LongPathsEnabled	RegQueryValue	882		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\FileSystem	RegCloseKey	883		
pingsender....	7976	C:\Program Files (x86)\Mozilla Firefox	CreateFile	884		
pingsender....	7976		Thread Create	885		
pingsender....	7976		Thread Create	886		
pingsender....	7976	C:\Windows\SysWOW64\ucrtbase.dll	Load Image	887		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	888		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegOpenKey	889		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager	RegSetInfoKey	890		
pingsender....	7976	HKLM\System\CurrentControlSet\Control\Session Manager\SafeDllSearchMode	RegQueryValue	891		
Showing 4,976 of 1,855,693 events (0.26%)					Backed by virtual memory	

Figure 35 contains the filters used to effectively block out all unnecessary processes which allowed us to generate figures 23 to 34.

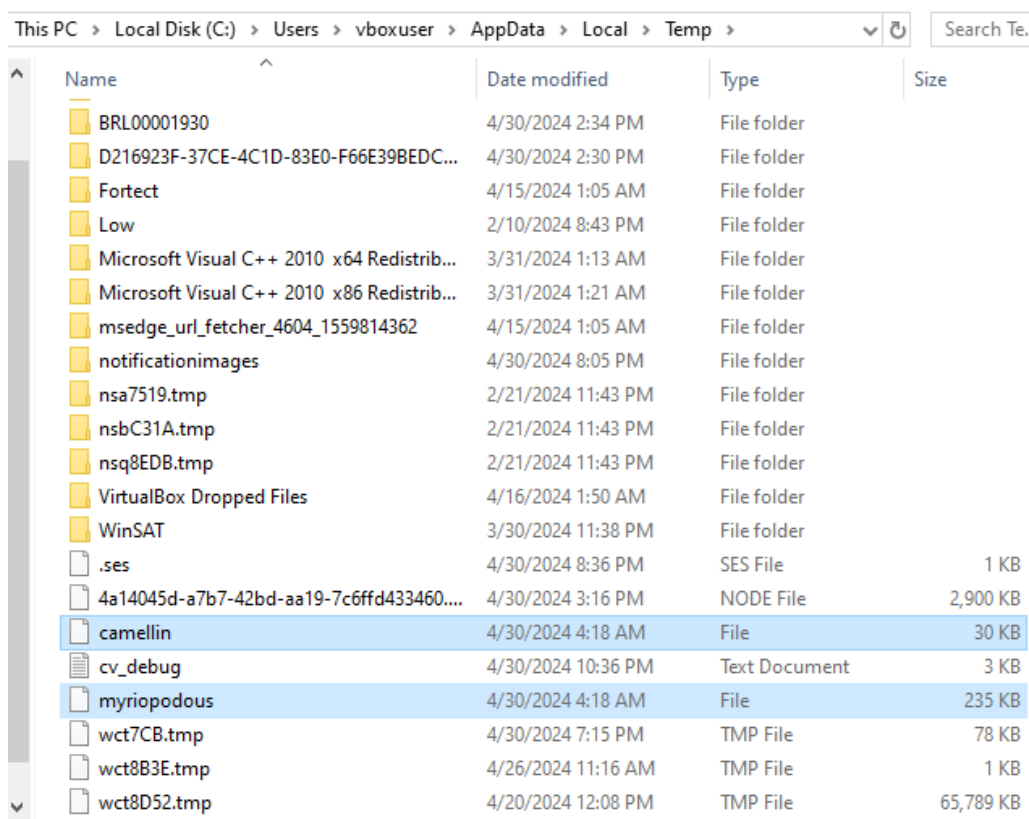
Figure 35.



Figures 23 to 31 show us the chain of operations executed after running the executable. In Figures 31 and 32 we finally see a reference to files. Specifically, two files called “myriopodous” and “camellin” located in “\AppData\Local\Temp”. The assumption is that the long list of executed operations is connected to these files which were created and written to. These files are likely to contain some sensitive user data or further instructions for the malware.

Now remember, we saw these files mentioned in VirusTotal in Figure 14 in the “Dropped Files” section. With only “camellin” being considered malicious, further solidifying the legitimacy of our findings. As we can see in Figure 36, the two files are present but in an unknown format. Furthermore, if we open them, they are in undecipherable encrypted format.

Figure 36.



Name	Date modified	Type	Size
BRL00001930	4/30/2024 2:34 PM	File folder	
D216923F-37CE-4C1D-83E0-F66E39BEDC...	4/30/2024 2:30 PM	File folder	
Fortect	4/15/2024 1:05 AM	File folder	
Low	2/10/2024 8:43 PM	File folder	
Microsoft Visual C++ 2010 x64 Redistrib...	3/31/2024 1:13 AM	File folder	
Microsoft Visual C++ 2010 x86 Redistrib...	3/31/2024 1:21 AM	File folder	
msedge_url_fetcher_4604_1559814362	4/15/2024 1:05 AM	File folder	
notificationimages	4/30/2024 8:05 PM	File folder	
nsa7519.tmp	2/21/2024 11:43 PM	File folder	
nsbC31A.tmp	2/21/2024 11:43 PM	File folder	
nsq8EDB.tmp	2/21/2024 11:43 PM	File folder	
VirtualBox Dropped Files	4/16/2024 1:50 AM	File folder	
WinSAT	3/30/2024 11:38 PM	File folder	
.ses	4/30/2024 8:36 PM	SES File	1 KB
4a14045d-a7b7-42bd-aa19-7c6ffd433460....	4/30/2024 3:16 PM	NODE File	2,900 KB
camellin	4/30/2024 4:18 AM	File	30 KB
cv_debug	4/30/2024 10:36 PM	Text Document	3 KB
myriopodous	4/30/2024 4:18 AM	File	235 KB
wct7CB.tmp	4/30/2024 7:15 PM	TMP File	78 KB
wct8B3E.tmp	4/26/2024 11:16 AM	TMP File	1 KB
wct8D52.tmp	4/20/2024 12:08 PM	TMP File	65,789 KB

Figure 37.

[illegible]

We also need to check if a secondary (backup) FQDN is present in the mechanics of the malware. To do this we'll set the # of NXDOMAINs to 1. As we can see in Figure 38, there isn't any secondary FQDN. Something fun to notice is a Windows mechanism is a call being made to "check.appexe.microsoft.com" right after the malware is executed as seen in Figure 38.

Figure 38.

The screenshot shows the ApateDNS application window. It has two tabs: "Capture Window" and "DNS Hex View". The "Capture Window" tab is active, displaying a table of DNS requests and responses. Below the table is a log of server operations. At the bottom, there are configuration fields for "DNS Reply IP (Default: Current Gateway/DNS)", "# of NXDOMAIN's", and "Selected Interface".

Time	Domain Requested	DNS Returned
23:23:48	array502.prod.do.dsp.mp.microsoft.com	FOUND
23:23:55	disc501.prod.do.dsp.mp.microsoft.com	FOUND
23:23:57	geover.prod.do.dsp.mp.microsoft.com	FOUND
23:24:36	disc501.prod.do.dsp.mp.microsoft.com	FOUND
23:24:38	kv501.prod.do.dsp.mp.microsoft.com	FOUND
23:24:52	disc501.prod.do.dsp.mp.microsoft.com	FOUND
23:25:21	wpad.student.rit.edu	NXDOMAIN
23:25:38	checkappexec.microsoft.com	NXDOMAIN
23:25:42	mail.cmcapama.top	NXDOMAIN
23:25:46	disc501.prod.do.dsp.mp.microsoft.com	NXDOMAIN
23:25:49	disc501.prod.do.dsp.mp.microsoft.com	FOUND

[+] Using 127.0.0.1 as return DNS IP!
[+] DNS set to 127.0.0.1 on Intel(R) PRO/1000 MT Desktop Adapter.
[+] Sending 1 NXDOMAIN replies to clients
[+] Server started at 23:25:19 successfully.
[+] Stopping Server...
[+] DHCP detected, setting DNS back to DHCP.
[+] DNS Restored.
[+] Interfaces list has been refreshed.

DNS Reply IP (Default: Current Gateway/DNS):

of NXDOMAIN's:

Selected Interface:

Additionally, after we set our ApateDNS configuration, we can see these results in Wireshark when we observe DNS and ICMP packets as seen in Figure 39 with red arrows.

Figure 39.

*Adapter for loopback traffic capture

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help

DNS || icmp

No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000	127.0.0.1	239.255.255.250	SSDP	169	M-SEARCH * HTTP/1.1
2	0.206200	10.0.2.15	239.255.255.250	SSDP	207	M-SEARCH * HTTP/1.1
3	1.221109	10.0.2.15	239.255.255.250	SSDP	207	M-SEARCH * HTTP/1.1
4	2.225506	10.0.2.15	239.255.255.250	SSDP	207	M-SEARCH * HTTP/1.1
5	2.999533	127.0.0.1	239.255.255.250	SSDP	169	M-SEARCH * HTTP/1.1
6	3.237608	10.0.2.15	239.255.255.250	SSDP	207	M-SEARCH * HTTP/1.1
7	8.952173	127.0.0.1	127.0.0.1	DNS	87	Standard query 0xe6a9 A array502.prod.do.dsp.mp.microsoft.com
8	8.952508	127.0.0.1	127.0.0.1	DNS	103	Standard query response 0xe6a9 A array502.prod.do.dsp.mp.microsoft.com A 127.0.0.1
9	8.953187	127.0.0.1	127.0.0.1	TCP	56	50256 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
10	8.953198	127.0.0.1	127.0.0.1	TCP	44	443 → 50256 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
11	9.455669	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50256 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
12	9.455712	127.0.0.1	127.0.0.1	TCP	44	443 → 50256 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
13	9.973698	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50256 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
14	9.973719	127.0.0.1	127.0.0.1	TCP	44	443 → 50256 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
15	10.308712	127.0.0.1	127.0.0.1	DNS	67	Standard query 0xe53b A mail.cmcapama.top
16	10.308912	127.0.0.1	127.0.0.1	DNS	83	Standard query response 0xe53b A mail.cmcapama.top A 127.0.0.1
17	10.314068	127.0.0.1	127.0.0.1	TCP	56	50257 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
18	10.314081	127.0.0.1	127.0.0.1	TCP	44	587 → 50257 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
19	10.482431	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50256 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
20	10.482451	127.0.0.1	127.0.0.1	TCP	44	443 → 50256 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
21	10.833013	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50257 → 587 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
22	10.833038	127.0.0.1	127.0.0.1	TCP	44	587 → 50257 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
23	11.003631	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50256 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
24	11.003653	127.0.0.1	127.0.0.1	TCP	44	443 → 50256 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
25	11.025016	127.0.0.1	127.0.0.1	DNS	86	Standard query 0xe4d5 A disc501.prod.do.dsp.mp.microsoft.com
26	11.025341	127.0.0.1	127.0.0.1	DNS	102	Standard query response 0xe4d5 A disc501.prod.do.dsp.mp.microsoft.com A 127.0.0.1
27	11.026055	127.0.0.1	127.0.0.1	TCP	56	50258 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
28	11.026072	127.0.0.1	127.0.0.1	TCP	44	443 → 50258 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
29	11.347495	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50257 → 587 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
30	11.347519	127.0.0.1	127.0.0.1	TCP	44	587 → 50257 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
31	11.533964	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50258 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
32	11.533984	127.0.0.1	127.0.0.1	TCP	44	443 → 50258 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
33	11.874288	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50257 → 587 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
34	11.874311	127.0.0.1	127.0.0.1	TCP	44	587 → 50257 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0
35	12.045591	127.0.0.1	127.0.0.1	TCP	56	[TCP Port numbers reused] 50258 → 443 [SYN] Seq=0 Win=65535 Len=0 MSS=65495 WS=256 SACK_PERM
36	12.045611	127.0.0.1	127.0.0.1	TCP	44	443 → 50258 [RST, ACK] Seq=1 Ack=1 Win=0 Len=0

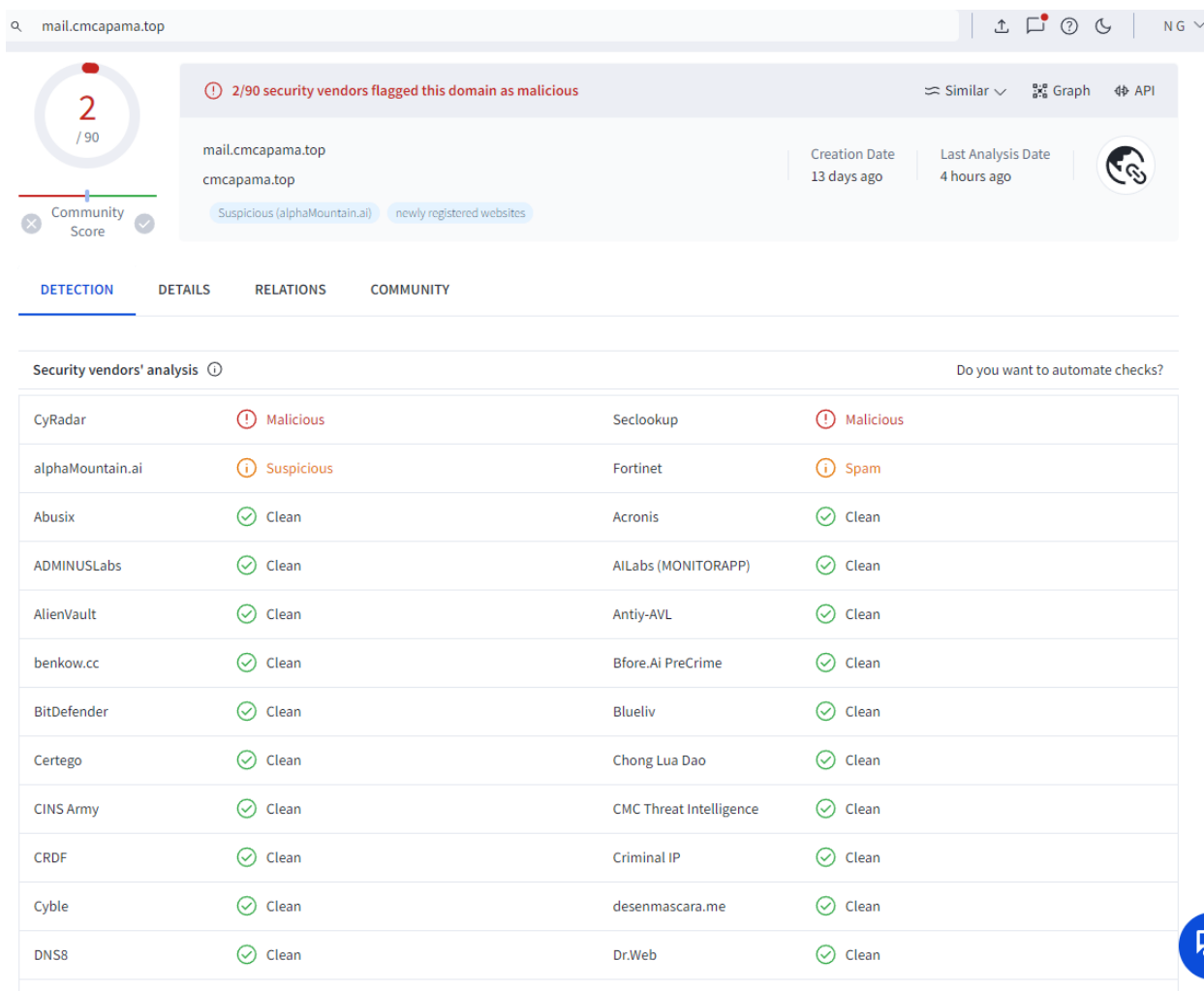
> Frame 15: 67 bytes on wire (536 bits), 67 bytes captured (536 bits) on interface \Device\NPF{...}\{...}\00000000-0000-0000-0000-00000000
> Null/Loopback
> Internet Protocol Version 4, Src: 127.0.0.1, Dst: 127.0.0.1
> User Datagram Protocol, Src Port: 57470, Dst Port: 53
> Domain Name System (query)

0000 02 00 00 00 45 00 00 3f 3c 22 00 00 00 11 00 00E...? <".....
0010 7f 00 00 01 7f 00 00 01 e0 7e 00 35 00 2b 4f 50~5+OP
0020 e5 3b 01 00 00 01 00 00 00 00 00 00 04 6d 61 69mai
0030 6c 08 63 6d 63 61 70 61 6d 61 03 74 6f 70 00 00 l'cmcapa ma-to-..
0040 01 00 01

AgentTesla: Quickly revisiting VirusTotal to check our results.

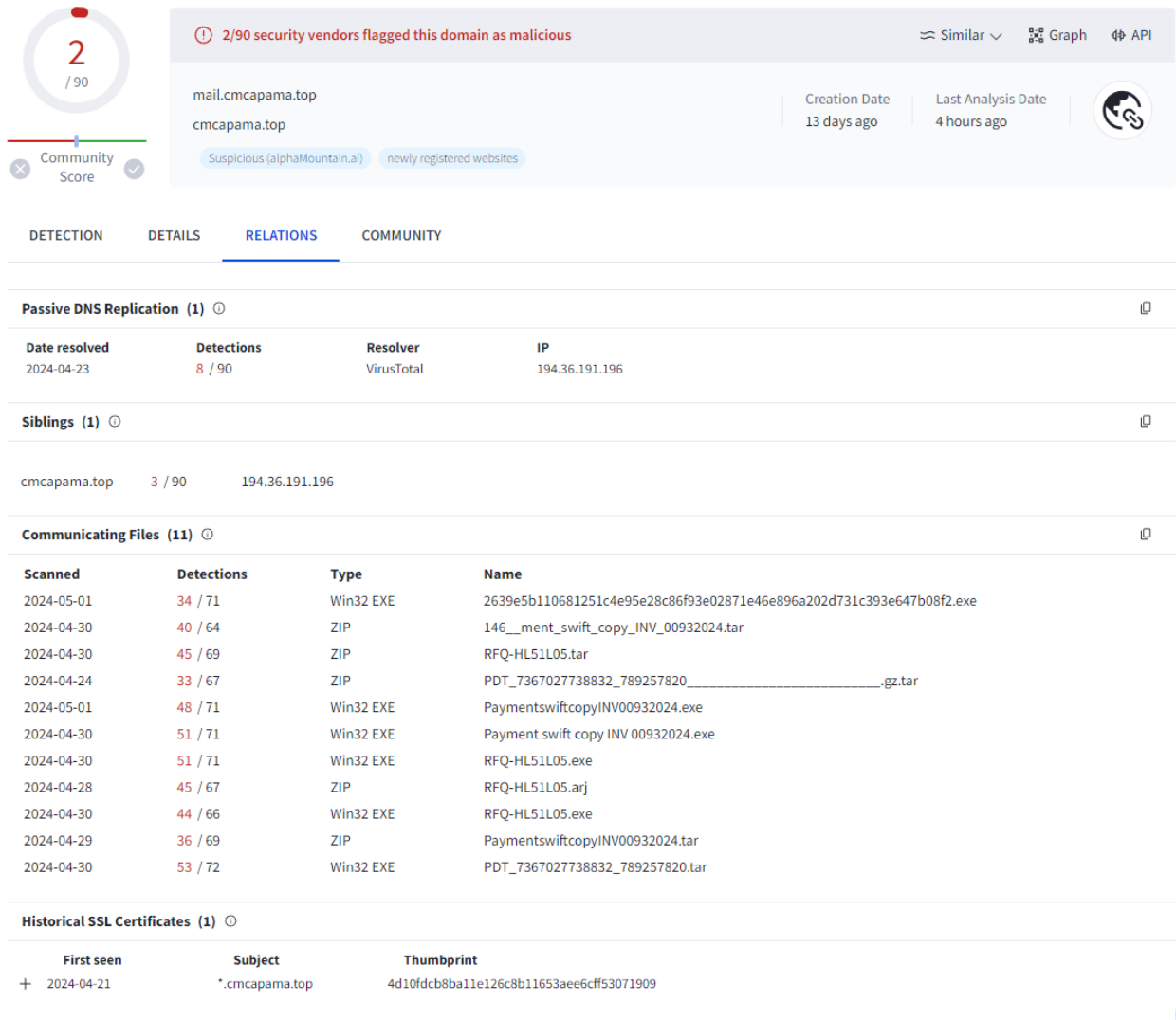
This section will serve to analyze the FQDN, its associated IP address, and the other malicious executables that connect to the same addresses (likely other variations of this malware strain derived from Agent Tesla). Figure 40 shows us all the other executables connecting to this FQDN. Amusingly, all of the executables are designated across the board as malware. However, the FQDN that this malware connects to, is only designated as malicious by two vendors. Again, indicative of a new strain of malware with a new Command and Control server not yet fully discovered.

Figure 40.



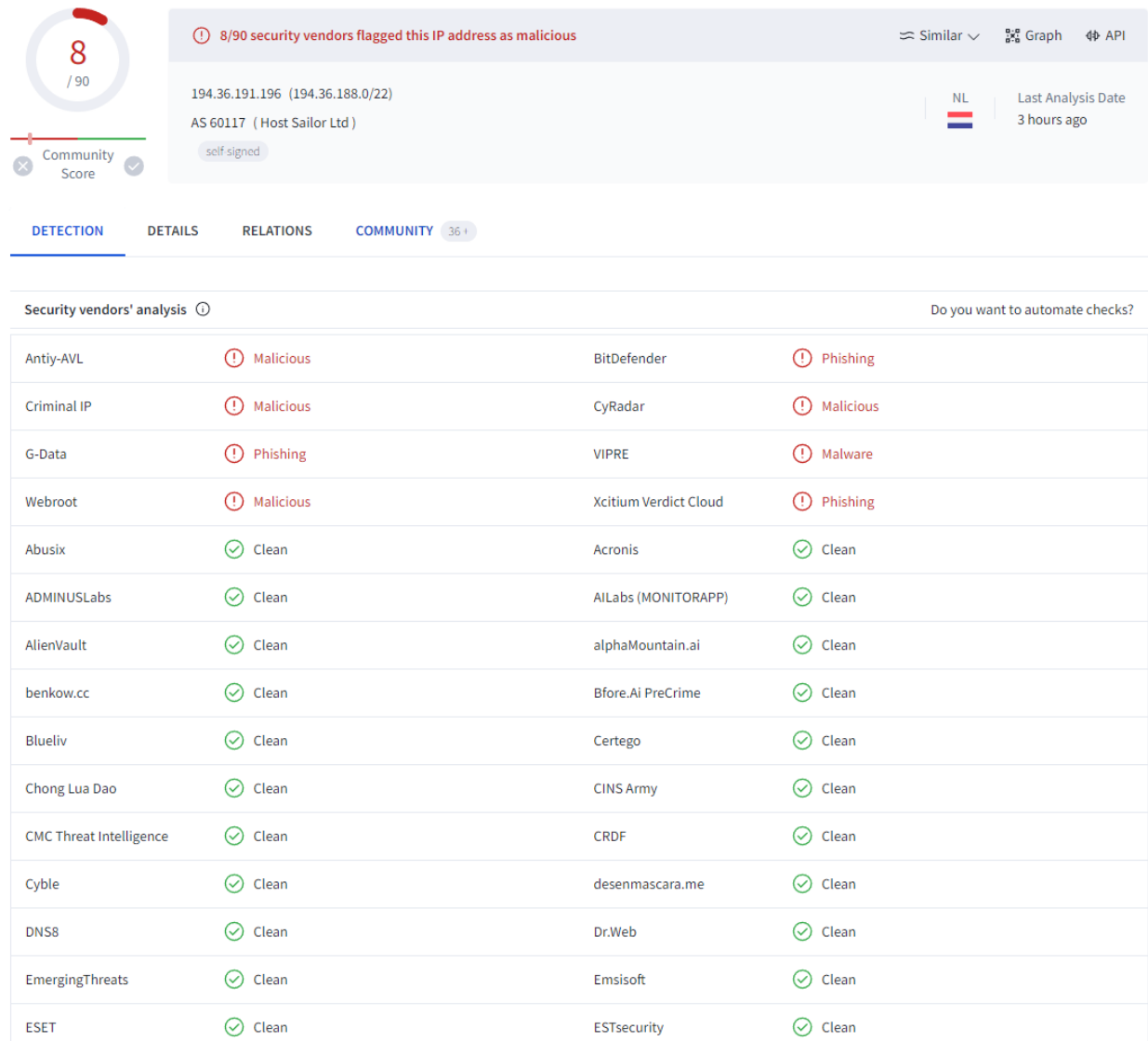
In Figure 41 we see some of the executable names that connect to this FQDN, one of them being INV 00932024.exe, if we recall this appeared in Figure 11 as an alias to our malware.

Figure 41.



Finally, we can also see that the associated IP address is flagged in Figure 42. However, again we see only a small number of vendors designating this address as malicious.

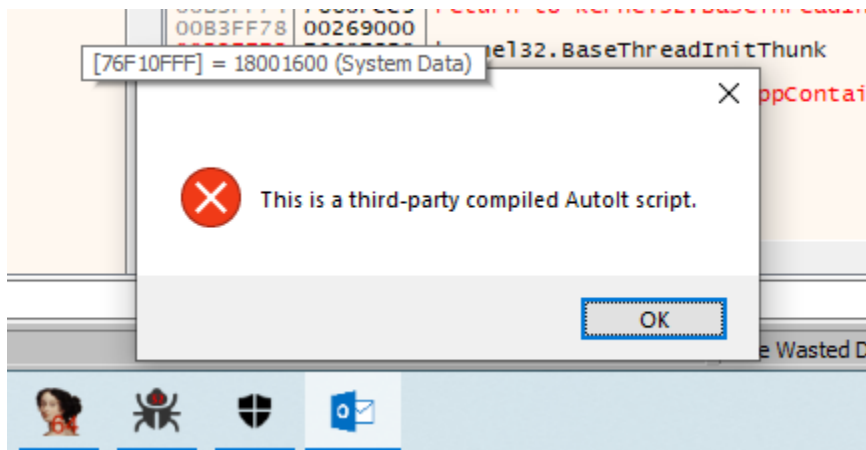
Figure 42.



AgentTesla: x64 Debugger & IDA.

During disassembly and debugging I started to have a hard time. I noticed some weird things occurring. During debugging, I received a weird error that never occurred when simply running the malware as shown in Figure 43.

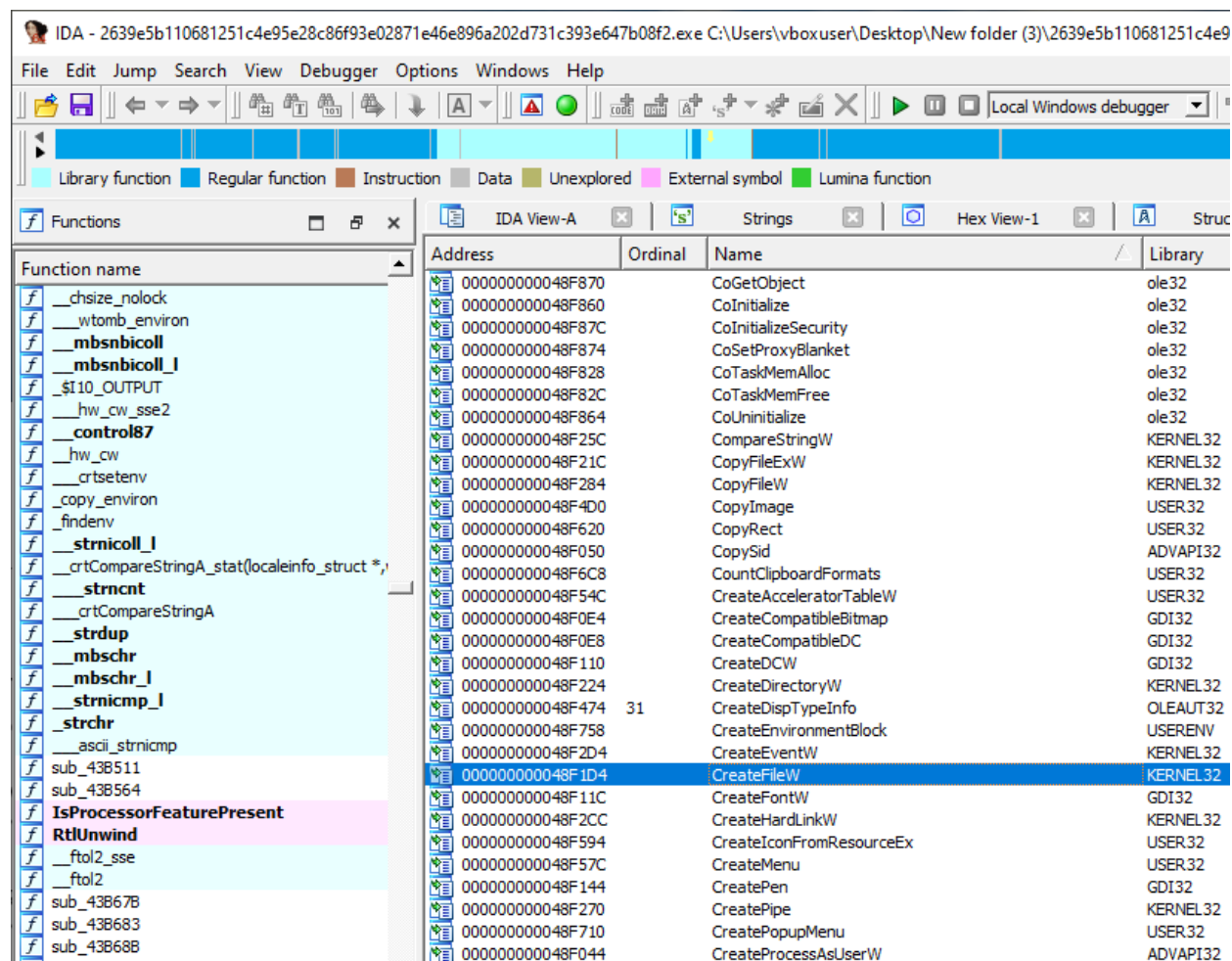
Figure 43.



This got me thinking that maybe I had some internal Windows safeguard mechanism enabled. So, I disabled every single option, and sub-option in Virus and threat protection, App & browser control, and Device Security. None of which changed the behavior of the debugged malware.

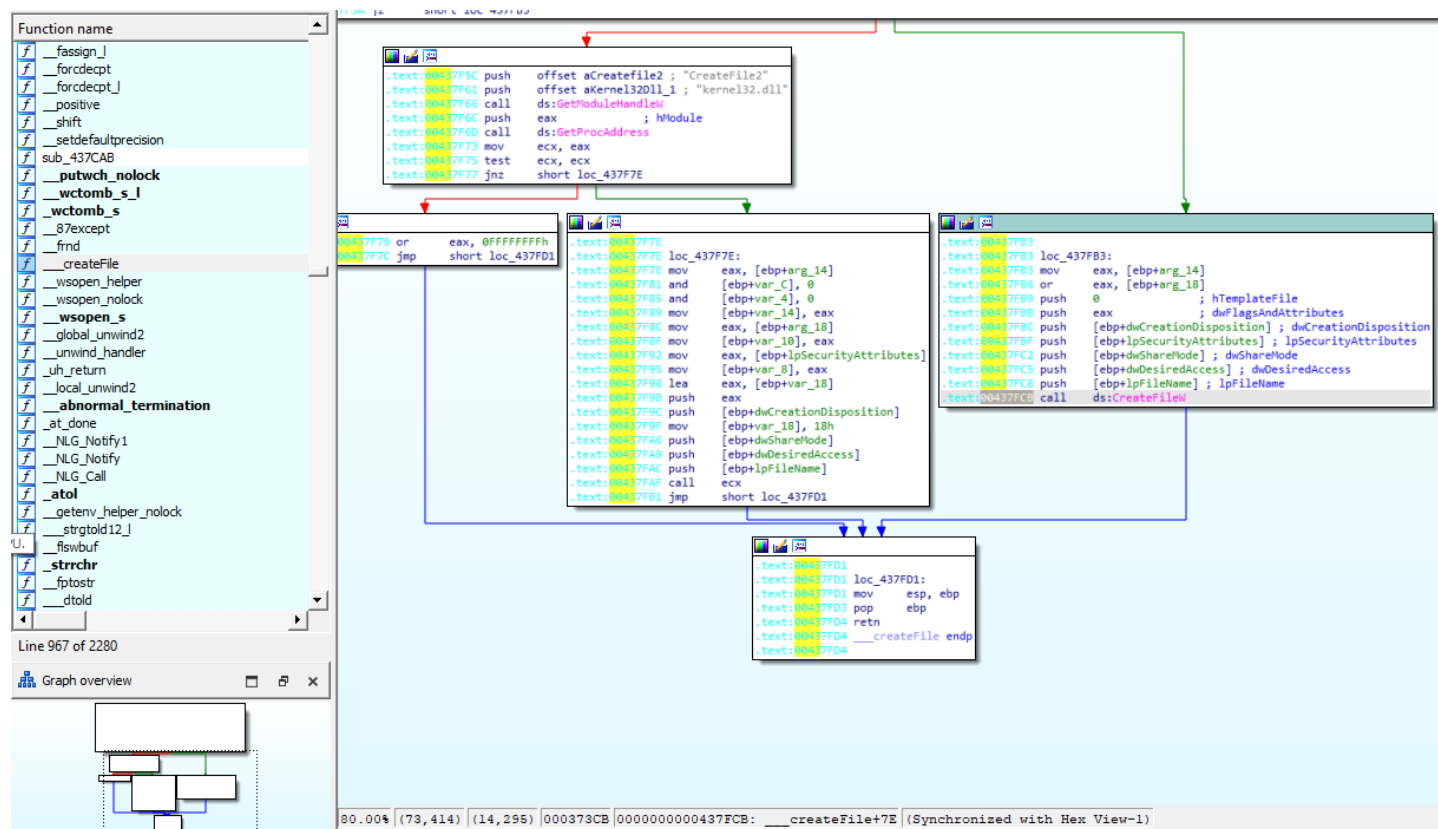
I also noticed that I would not be able to reach certain functions found during disassembly while debugging, such as the "CreateFileW" function as shown in Figure 44. This function was chosen based on the fact that this malware is a portable executable, the first action that it should logically attempt is to create another file or alter registry values possibly for persistence or redundancy.

Figure 44.

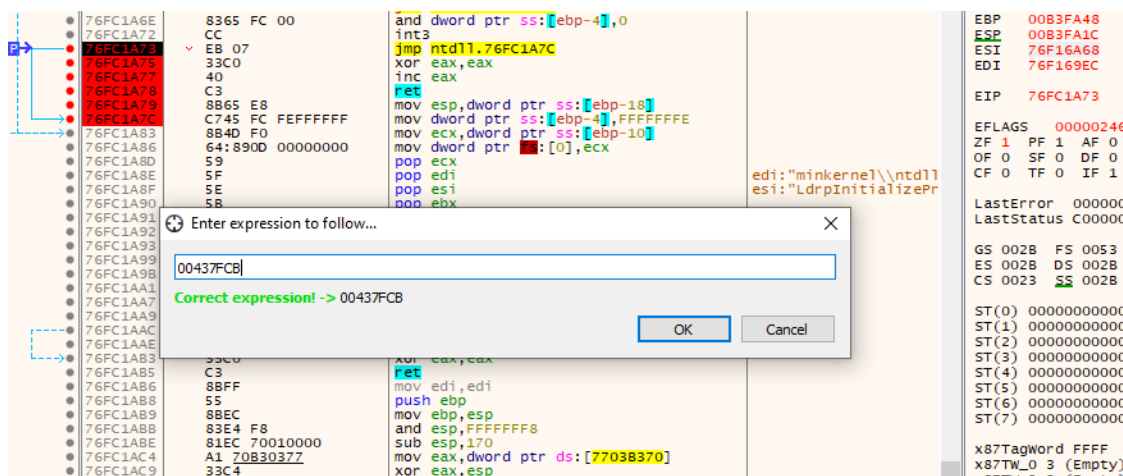


When we cross-reference this built-in Windows function, we can see it being used in another function called “__createFile” as seen in Figure 45. This is where some of the issues started appearing. Upon investigation, we found that the address responsible for calling this function is “00437FCB”, as highlighted in gray in Figure 45 as well.

Figure 45.

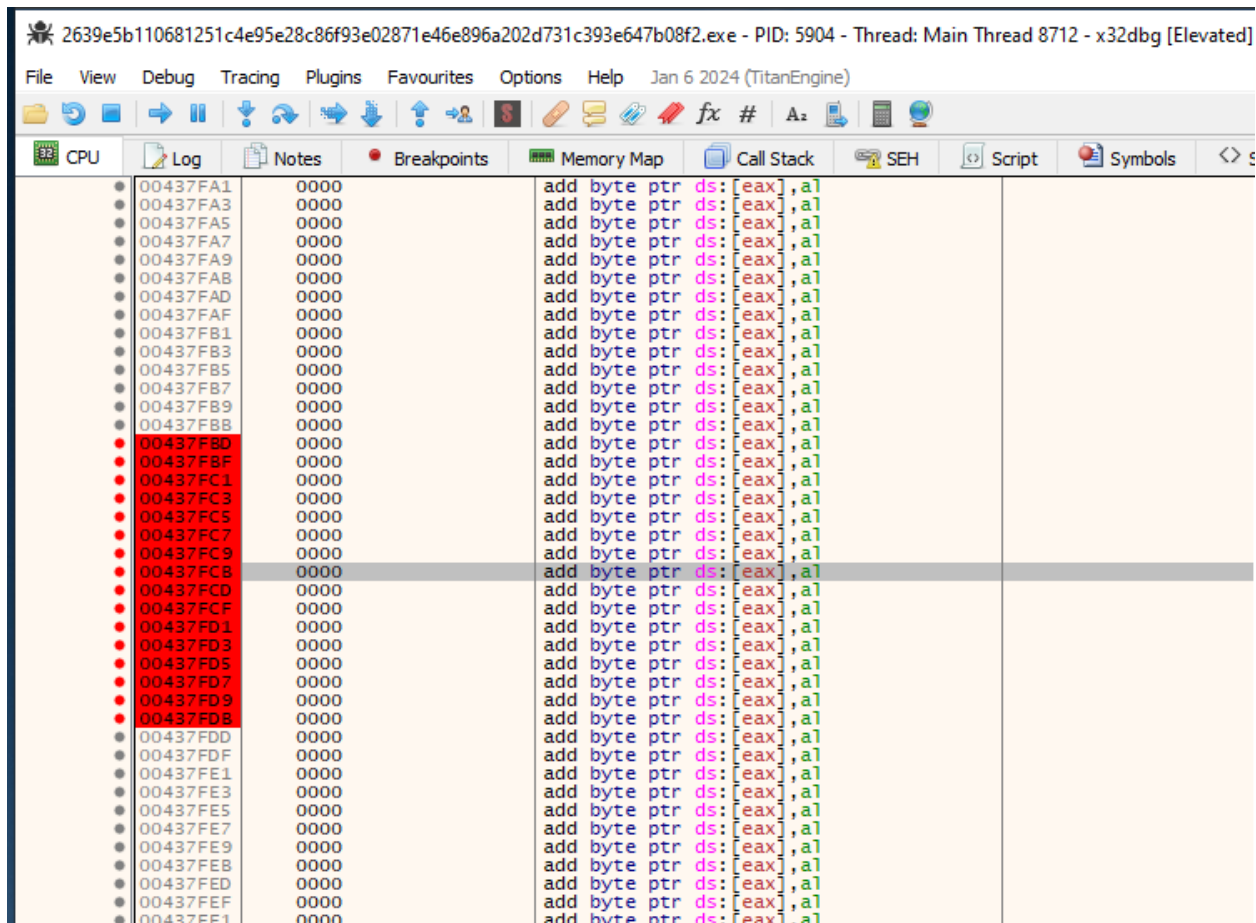


Upon attempting to reach the file in x64 Debugger, I noticed the address existed as shown in Figure 46, which was a good sign.



However, when we examine the instructions, we see some very unusual entries as seen in Figure 47. Not only are ALL of the instructions related to the “CreateFileW” function the same “add byte ptr ds:[eax],al”. The debugger never even reaches these instructions anyway!

Figure 48.



This got me thinking that maybe there are some safeguards built into the malware to protect itself from being disassembled or altering its behavior in a Virtual Machine as we learned in our course. And unfortunately, after some digging around on Google searching for “agent tesla in virtual machine”, I found two reasons why the malware was behaving irregularly.

As pointed out by Acronis Detection and Response, the malware checks whether it is running on a Virtual Machine or not by querying registry values (Acronis, Inc.). This is exactly what we saw happening in Figures 23 through 34, now making more sense. Secondly, the malware has safeguards built into it so that it doesn't allow for clean debugging. It employs a "crypter" called AspireCrypt, which disrupts our process of debugging (Acronis, Inc.).

Regardless, some of our findings seem consistent with other vendors. The Agent Tesla variation mentioned by Acronis, seems to also point to a mail server just like ours did, just with a different address. This shows us that the source code for this malware must have been leaked, made public, or preconfigured for customers since the behavior is identical with only changes being made to addresses.

Summary

Agent Tesla is an extremely powerful tool as analyzed in this report. It is a Remote Access Trojan (RAT) that has many capabilities. Some of these include but are not limited to keylogging, stealing clipboard data, screenshot capturing, and webcam access. It encrypts its stolen data so that there will be no way for users to decipher what was stolen. The malware uses a mail server as its Command and Control, however, these are different from other strains of Agent Tesla.

Both variations that we saw of Agent Tesla (Acronis and ours) employ techniques for identifying whether Agent Tesla is being run on a Virtual Machine by querying registry data. As reported by Acronis, we saw that a Crypter is also utilized in this malware, making it highly difficult to detect and analyze.

Removal Recommendations:

If the malware was not detected by your anti-virus or the built-in Windows Defender and you suspect you have been infected with AgentTesla then I would highly suggest going with the zero-trust approach. With this approach, a clean install of windows should suffice. Additionally, it is safe to assume any passwords stored or typed recently on the machine are compromised (safe to assume browser passwords as well). Thus, rotating passwords for all major accounts will be crucial for security. However, should this not be a valid option, here are some steps you could use to potentially remove the malware:

1. Isolate the Infected System: Disconnect the PC from the internet to prevent its spread and communication to its command server.
2. Use good Antivirus Software: Using MalwareBytes Antimalware should be a good pairing to use with the built-in Windows Defender antivirus.
3. System Restore: If a restore point was made before running the malware, then consider reverting to that point.
4. Update: Using Windows Update in addition to any other anti-virus updates should improve detection and removal of the malware.
5. Change Passwords: Change all system, application, and website-related passwords while also enabling two-factor authentication (2FA).
6. Monitor System Traffic: As we saw, should a connection be made to a similar mail FQDN, then suffice to say that the malware is still present on the PC.

References

MalwareBazaar:

abuse.ch. "Malware Sample Entry for SHA256:

2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2." *MalwareBazaar*, 2024,

<https://bazaar.abuse.ch/sample/2639e5b110681251c4e95e28c86f93e02871e46e896a202d731c393e647b08f2/> . Accessed 30 Apr. 2024.

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Acronis Cyber Protection, 2024, <https://kb.acronis.com/content/68881> . Accessed 30 Apr. 2024.